



Home broadband and human capital formation

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ABSTRACT

Using administrative data from England, we estimate the effect of home broadband speed on student-level value-added test scores. Our primary estimation leverages jumps in connection quality between close neighbors across thousands of invisible telephone exchange station catchment area boundaries. We find that home broadband speed has positive effects on general measures of human capital, with these effects concentrated among high-ability students and those not eligible for free school meals. These positive outcomes result from more education-oriented internet use. However, these effects are observed only for students who attend schools with faster broadband connections. Our study documents a complementarity between home and school technology in relation to general measures of human capital. Policies introducing new learning technologies should take this complementarity into account.

1. Introduction

Technological advances in information and communication technology (ICT) have transformed the education landscape over the past two decades. Currently, the educational technology (*edutech* or *edtech*) industry is booming, with a wide consensus among researchers, policymakers, educators, and parents on the importance of integrating these technologies into learning environments. However, there is no consensus on their net impact on educational outcomes.² Although improved home internet quality can improve learning productivity and expand educational opportunities, it may also lead to unproductive distractions, making its net effect on the formation of human capital of students ambiguous (Bulman & Fairlie, 2016).

The central contribution of this paper is to isolate the impacts of home broadband quality, captured by speed, on general measures of human capital for teenagers, and to examine how they interact with individual and school-level factors.³ Our key finding, based on administrative data from England from 2005–2010 combined with granular spatial regression discontinuity design, is that increases in home broadband speed improve student test scores aligned with the national curriculum. Our results are particularly relevant because the literature has shown that better standardized test scores, as a measure of human capital, are linked to knowledge accumulation and economic development (Angrist et al., 2021).⁴

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² We discuss the related literature in detail below.³ Our study does not assess the impact of accessibility to broadband on student performance, but of variations in speed for a fixed technology, for which the evidence is much more scarce. For other studies looking at the impact of access, see the literature review⁴ The accumulation of human capital, i.e., the stock of skills, traits, and knowledge that an individual possesses (Burgess, 2016), is the key to growth, employment and earnings (Barro, 2001; Becker, 1962; Mincer, 1974), with educational outcomes such as cognitive skills serving as reliable proxies (Hanushek & Woessmann, 2012, 2015).

Our heterogeneity analysis reveals that these positive effects are driven by wealthier students from non-free school meal backgrounds. Our research design carefully controls for school-level factors, ensuring that our estimates are causally linked only to the home environment. In the case of England, there is no direct mapping between students' home addresses and schools attended. This specific feature of the English secondary education system allows us to explore interactions between school and home broadband quality. We find that the positive effects of home broadband speed only occur when students attend schools with faster broadband access. We present empirical evidence supporting a productivity-enhancing mechanism, demonstrating positive effects from utilizing faster broadband for educational purposes at home, without any significant impact on broadband take-up or participation in distracting activities. The implication of our findings is that access to high-quality home broadband matters to human capital and has broad economic implications.

Obtaining causal estimates of the impact of home internet quality on test scores is not without its challenges. First, there is *active* selection, i.e., household choices about broadband packages. More affluent or better educated parents may have a higher willingness to pay for faster broadband packages, inducing spurious correlations between teenage test scores and observed broadband speeds at home. To overcome active selection, this study exploits the differences in available (rather than realized) broadband speed.⁵ The available broadband speed at home decays with its distance from the local telephone exchange (LE), which provides internet and telephone services to the addresses connected to it. Hence, we exploit this technological feature of DSL broadband that generates differences in available broadband speed within very small neighborhoods, allowing us to address *passive* selection concerns, i.e., results unintentionally driven by technological differences correlated with neighborhood characteristics. We estimate effects using a spatial regression discontinuity (SRD) design at a granular level, described in detail in Section 3.

Multiple sets of administrative microdata meet the extensive data requirements for our empirical strategy. We use standardized and externally marked test scores for 14 and 16-year-olds in England's national Key Stage 3 (KS3) and KS4 tests from 2005 to 2010, along with rich background information, including pre-Internet test scores. We georeference student residences and schools at the most granular level—English postcodes, roughly equivalent to blocks of 15 addresses. We use data from the telecom network, including the location of approximately 3,900 telephone exchange stations and their postcode assignments, to identify invisible boundaries in exchange connections. We complement this with a rich vector of georeferenced control variables, including local amenities and property transaction values to validate our empirical design. Lastly, using the British Household Panel Survey (BHPS) for 2007, we investigate mechanisms driving broadband speed's impact on educational achievement, analyzing data on internet access, computer use, and household characteristics. Although this survey covers slightly older teenagers, due to data limitations, the evidence is still informative about the channels driving our results, which are completely novel in the impact of technology on education literature.

Our spatial regression discontinuity analysis shows that the available broadband speed positively affects externally marked test scores that measure learning progress against the national curriculum. Section 4 presents robustness checks to validate our strategy, with results remaining stable upon controlling for school-specific broadband features and including school or school-by-year fixed effects. This stability implies that school or teacher inputs do not change discontinuously across invisible technological broadband speed boundaries. Heterogeneity analysis indicates that positive effects arise only when pupils

attend schools with fast broadband and if these schools receive early DSL upgrades. The results are driven by students from more advantaged socioeconomic backgrounds and higher abilities, amplifying preexisting differences and highlighting important complementarities in learning technologies. These effects persist two years later when students are 16 years old.

The “proximity to local exchange” estimates are intent-to-treat (ITT) impacts of available faster broadband, relevant to understanding the distance-test score relationship. In Section 5, we use additional datasets to convert SRD estimates into broadband speed effects by estimating speeds for our study period with technology-adjusted postcode-level speed data. Consistent with existing studies for this time period (Ahlfeldt et al., 2017), we estimate that for each additional 100 m closer to the connected LE station, the local average speed increases by 0.089 Mbit/s. This implies that a speed increase of 1 Mbit/s (20% over baseline) increases the test scores by 1.37 percentile points, equivalent to about 5% of a standard deviation in the national test scores, which can be understood as a cumulative effect over six years. Assuming linear effects over time, each year of faster internet exposure boosts test scores by 0.8% of a standard deviation, with comparisons to other human capital interventions discussed in the conclusion.

In Section 6, we explore the mechanisms behind these results with four findings: households on the fast side do not experience a higher internet or broadband take-up; there is no effect on distracting activities such as video games; teenagers aged 15–17 use computers more for education than older family members, and those on the fast side are 27% more likely to use them for educational work. Interestingly, the positive effects of faster broadband speed on educational use are not present in young adults. For teenagers, we interpret these as reflecting a more education-focused internet use combined with limited distraction.

Although it seems the use of superfast broadband in education might be ubiquitous, our results are still very relevant in the current digital era, as DSL broadband remains a driving force. The share of fiber connections in the OECD countries is around 40% in 2023, increased from 27% in 2019 and 12% in 2010, making it the most prominent internet technology, but mostly in very recent years. In 2021, DSL still accounts for approximately 22% of global broadband connections, equivalent to around 1.188 billion users (OECD, 2024). Furthermore, for existing DSL connections, even in the UK - a country with advanced infrastructure - the average ADSL speed as of March 2023 is only about 11 Mbit/s (OfCom, 2023). This is about twice as fast as the average speed ADSL speed that we observe in our study period, corresponding to comparable or even lower effective speeds today. This is because the speed requirements even for standard browsing have increased alongside, with more content-heavy webpages. For example, the default screen resolution has increased from 1024 × 768 in the 2000s to 1920 × 1080 (Full HD), resulting in more than twice the data requirements for displaying a picture in the same on-screen magnitude. In fact, many families lack broadband even in developed countries, with only 56% of low-income US families and 51% of UK families having subscriptions (Commission et al., 2021; Zuo, 2021). This underscores our estimates' external validity and suggests that our results on school-home technology complementarity for broadband are still relevant today. In the conclusion, we also relate our findings to the emerging literature on mobile phones and tablets and directions for future research.

Our paper joins a small but growing body linking home broadband technology to student test scores.^{6,7} Malamud et al. (2019) find no

⁵ This limits the time period for causal analysis to before the wide availability of fiber broadband, similar to much of the existing literature.

⁶ Table A.1 provides a comprehensive literature overview. Broadband's socioeconomic impacts include labor productivity and wages (Akerman et al., 2015), economic growth (Czernich et al., 2011), property market capitalization (Ahlfeldt et al., 2017), health choices (Amaral-Garcia et al., 2019), m/0258ge rates (Bellou, 2015), and political participation (Campante et al., 2018; Falck et al., 2014; Gavazza et al., 2019), sex crime (Bhuller et al., 2013), and social capital (Geraci et al., 2022).

significant effects of home internet access on achievement in Peru using a randomized trial providing laptops and internet in low-achieving schools. Our paper expands by examining a broader, older student population across all socio-economic levels in a developed country. Our results are driven by speed changes, not access. In another key study, [Dettling et al. \(2018\)](#) shows how high-speed internet access improves SAT scores and increases college applications in the US, driven by access over quality, with positive effects concentrated among high-SES students. We complement and extend these findings to administrative achievement tests assessing cognitive ability within the regular curriculum. Unlike [Dettling et al. \(2018\)](#), whose results are not due to GPA improvements, we find that academic performance gains are reflected in standardized test scores. A key insight of our outcome, the Key Stage 3 test, is its low-stakes nature, designed to assess progress without explicit online training, offering insights into the broader impact of home environments on human capital formation beyond ICT skills, test preparation or information about colleges.

Related literature examines home computer access's educational outcomes using quasi-experimental and randomized methods ([Beuermann et al., 2015](#); [Cristia et al., 2017](#); [Fairlie & Robinson, 2013](#); [Malamud & Pop-Eleches, 2011](#); [Vigdor et al., 2014](#)), often showing positive computer-access-related outcomes but little or negative academic effects. [Malamud and Pop-Eleches \(2011\)](#) find parental computer and homework rules mitigate negative low-SES impacts in Romania. We focus instead on intensive-margin broadband speed changes. Finally, studies also explore ICT and broadband at the school level ([Angrist & Lavy, 2002](#); [Belo et al., 2014](#); [Goolsbee & Guryan, 2006](#); [Lakdawala et al., 2023](#); [Machin et al., 2007](#); [Rouse & Krueger, 2004](#)). Connecting the two, [Belo et al. \(2016\)](#) shows that broadband at school can increase broadband usage at home. In this paper, we study the effects of home broadband on learning. This is a related but different question compared to effects at the school, where, for example, teachers are present and need to think about how to incorporate the technology into the learning process ([Falck et al., 2018](#)). We show that the quality of home broadband matters for a broad measure of human capital in a developed-country context.

2. Background, data and descriptive statistics

2.1. Broadband expansion in England

The empirical setting of this study is England over the time period 2005–2008, where the broadband market was already developed. The rollout of DSL broadband technology in England, as part of the United Kingdom (UK), started in the major urban centers at the beginning of the 2000s and proceeded rapidly. This process involved technological upgrades of the infrastructure of telephone LE stations – the same ones that provide telephony services to a number of connected premises around them – to allow them to offer broadband internet services through copper cable. By the end of 2004, 80% of the LE stations

had been equipped to provide broadband services, covering 97% of local residences, which could subscribe to receive broadband services at home. That year, 54% of households had an internet connection, of which 6.2 million (approximately 25%) were broadband. By the start of our estimation period in 2005, 99% of English addresses were connected to broadband-enabled telephone LE stations.

Even if most of the technological upgrades took place between 2000 and 2005, penetration rates were low in the first years (approximately 10% in 2003) and only started growing in 2004. By then, infrastructure was completely rolled out across space, and the take-up rate increased steadily. The broadband internet take-up rate rose from approximately 30% in 2005 to over 60% by 2008 (Eurostat). This increase in take-up was related to decreases in prices and changes in attitudes and internet content. In 2008, 84% of students used the internet, and among those, 90% used it for their homework ([Livingstone & Bober, 2005](#)). Due to this, and similarly to existing work in the UK context ([Nardotto et al., 2015](#)), we focus on the post-2005 period in our analysis. In this context, we can focus on the impact of broadband speed for a given state of technology and exploit very local variations in its quality.

The relatively fast broadband development in the UK also makes this period an interesting one to study from today's perspective. In 2007, more than half of UK homes had broadband access, with an average connection speed of 4.6 Mbit/s ([OfCom, 2009b](#)). While today the average internet speed is faster, the coverage and speed available to English households between 2005 and 2008 are comparable to the infrastructure currently available in large parts of the world. For comparison, in 2019, only approximately 16% of the world population had broadband access ([Roser et al., 2020](#)), with average connection speeds of 7.2 Mbit/s ([McKeay, 2017](#)). And even today, many families live without any broadband connection in developed countries, with only 56% and 51% of low-income families with broadband subscriptions in the US and UK, respectively ([Commission et al., 2021](#); [Zuo, 2021](#)).

Note that our period of analysis ends in 2008 for two reasons: (i) mobile broadband and cable/fiber internet technologies became more widespread in the UK after 2009, reducing the efficacy of our empirical approach, and (ii) the standardized exam that we use to measure the educational achievement of teenagers in this paper (the KS3 for students at age 14) was discontinued. The combination of the testing regime and the state of development of the broadband infrastructure in England in the period 2005–2008 offers a unique opportunity to study the effects of home broadband speed on student performance.

2.2. Data

2.2.1. Administrative student records

In the English educational system, student academic performance is assessed in national exams that are administered through externally marked tests. The English education curriculum is organized into four key stages (KSs). Compulsory education starts at age 6 and ends at age 16 with the fourth and final KS4 (the General Certificate of Secondary Education [GCSE] examinations). There are several reasons why the KS3 exam at age 14 is the most suitable for our analysis. First, the KS3 test makes all students sit the same set of exam papers and is externally marked and thus comparable across students and schools. Second, the test is low stakes, so there are no incentives for teachers or students that would drive a wedge between test scores and real achievement. Third, the test is finely graded (mostly zero to 100); therefore, in combination with our sample size, it is possible to detect even small effect sizes. Finally, all students are tested in the three main compulsory subjects: English, mathematics and science. Students have very limited options in choosing subjects or specializing according to interest or ability before the KS3, in stark contrast to the educational period before the KS4 test two years later. For the earlier KS1 and KS2 tests, children are too young to be using the internet much. In our analysis, we use the teacher-graded KS1 scores to control for individual-level ability.

⁷ This paper diverges from an earlier related working paper by [Faber et al. \(2015\)](#) in several ways. Following referee and editor suggestions, it now focuses on the mature broadband market years in England (2005–2008/10) without other confounding technology (see Section 2.1). Treatment is redefined as continuous broadband speed changes, not DSL access/no-access, and the time period allows for pupil-specific (pre-broadband area) KS3-KS1 value-added regressions accounting for individual heterogeneity, enhancing precision (see Section 2.2.1). Aggregation necessitating fuzzy spatial RD further increases precision (see Section 3.3). This shifts a formerly imprecise zero finding to our significant non-zero effects (Section 4.3, which also re-estimates the less precise zero estimates for our study period in Table A.3). The previous version also lacked direct evidence on the test score-driving mechanisms and speed-student performance estimates using the Wald-estimate technique. Moreover, this work carefully examines possible interactions with school-level technology.

For the main analysis, we employ administrative data with information on the universe of students enrolled in English state schools (approximately 95% of pupils) who took the KS3 test from 2005–2008 in England. These data are supplied by the Department for Education (DfE). To match the student information with the telecom network data that we describe below, we first use the restricted-access version of the National Pupil Database (NPD), from which we extract the full residential postcode for each registered student in a given year. British postcodes are associated with a small number of addresses (15 on average) and in denser areas usually correspond to housing blocks. In the second step, we use the unique student identifiers to link their residential information to individual test score results, which are also provided as part of the NPD.⁸

Following the education literature, we transform these scores into percentile ranks for each test and cohort, i.e., separately by year-subject. These subject percentiles are then added into a total score, which we percentilize to obtain an average total score ranging from 1 to 100. We conduct this transformation to make our results comparable to other countries' national exams as well as across cohorts/subjects. Transforming raw scores into percentile ranks has the goal of keeping the ordinal information in the outcome variable and removing the cardinal differences between units of interest, which might be driven by the setup of any particular exam paper, for instance.

We use additional data from the DfE NPD for each pupil in our KS3 2005–2008 sample and collect information on their KS1 test scores (taken at age 7). For this sample, this corresponds to tests taken during 1998–2001, when most of the rollout of broadband internet had not yet taken place and the level of broadband take-up was essentially zero. In contrast with the KS3, this test is marked by the schools, is only available for the subjects of mathematics and English, and is graded on a coarse scale. However, adding this information to our empirical models allows us to estimate individual-level value-added results, controlling for pupil-specific time-invariant ability and background characteristics, which in turn improve the precision of the estimates and the explanatory power of the models. We also obtain information on the location, size and type of the school that the pupils attend, which we use to construct school-level controls and, for some specifications, school fixed effects.

In addition to test scores, the administrative data give us access to a series of observable student characteristics, such as gender, ethnicity, and student eligibility for a free school meal (FSME), which is a common proxy for family income. We exploit these data at two scales: to construct individual-level controls and to calculate postcode-year-specific demographics based on the population of pupils of all ages, which we also use as local area control variables in the regressions.

2.2.2. Average house prices and area socioeconomic characteristics

We use a number of additional datasets to improve precision and to validate our approach. First and foremost, we use transaction-level data on property sales in England over the estimation period. The data are administrative records from the England and Wales Land Registry, covering all property transactions over this period. We use the reported property address information to link these property transaction values to individual residential postcodes. The postcode-year averages are based on several million individual property transactions that occurred in England over the period 2005–2008. Local house prices capitalize many desirable (and undesirable) local attributes and are likely to capture a large number of unobserved spatial characteristics of the areas.

Even though our empirical analysis is based on a spatial discontinuity design that compares only very proximate households, it could

⁸ The DfE formerly distinguished between the NPD and the Pupil Level Annual School Census (PLASC), which is now treated as part of the NPD. Note that no information is available on private schools, which enroll approximately 6%–7% of the English student population (Ryan & Sibiet, 2010).

Table 1
Summary statistics.

	All sample (1)	Baseline sample within 300 meters (2)
A. Outcome Variables		
Average Percentile Rank Score (Mean)	50.21 (28.67)	49.76 (28.66)
Average Percentile Rank Score in English	50.26 (28.59)	50.21 (28.51)
Average Percentile Rank Score in Maths	51.60 (28.03)	51.09 (28.05)
Average Percentile Rank Score in Science	51.50 (27.97)	50.84 (27.98)
B. Discontinuity Variables		
Distance to the Segment (Meters)	679.13 (547.1)	156.29 (79.1)
Distance to the LE Station (Meters)	1,511.96 (862.7)	1,866.97 (878.3)
Share on the “Fast” Side	0.54 (0.50)	0.51 (0.50)
Average “jump” (Meters)	763.36 (659.7)	930.29 (598.5)
C. Pupils & School Characteristics		
Distance to School (Meters)	2,602.5 (3,433.14)	2,413.35 (3,008.3)
White	0.838 (0.37)	0.785 (0.41)
Male	0.499 (0.50)	0.498 (0.50)
Free School Meal	0.141 (0.35)	0.155 (0.36)
Pre-KS3 Score (KS1)	44.45 (24.61)	43.98 (24.65)
Number of Schools	2,864	2,610
D. Area Socioeconomic Characteristics		
Share of White Pupils	0.825 (0.29)	0.769 (0.32)
Share of Free School Meal Pupils	0.151 (0.24)	0.165 (0.24)
Share of Community Schools	0.638 (0.48)	0.620 (0.48)
Average House Prices (Pounds)	193,092.6 (118,208.4)	190,596.8 (110,831.9)
Observations	1,115,594	183,892

Notes: This table shows descriptive statistics for the outcome variables (Panel A), treatment variables (Panel B), pupils and school characteristics (Panel C), and density and area socioeconomic (Panel D). The first column reports statistics for the whole sample of pupils and postcodes. The second column shows statistics for our baseline sample, which are pupils and postcodes located within 300 meters of the LE station boundary segment. Standard deviations are reported in parentheses.

still be the case that catchment area boundaries coincide with physical barriers such as roads or rivers and that either the slower or the faster side of the boundaries has a higher likelihood of hosting a given type of local amenity, the combination of which could lead to bias in the boundary effect. Using a GIS with detailed attribute data from the UK Ordnance Survey, the commercial real estate consultancy CBRE and the DfE, we compute euclidean distances between each English postcode and the following features: nearest school (primary or secondary), nearest road (class A, class B and motorways), nearest rail station (which captures centrality), nearest water body (river, stream, marsh or lake) and nearest supermarket. One of the major concerns is that *passive* endogeneity arises because local geography correlates with the location of the LE stations and of households. Taking these variables into account allows us to properly test whether observable geographic features are an endogeneity concern in our setting and ultimately control for these variables in our empirical specification to increase statistical precision.

Finally, we combine data from the Office for National Statistics Postcode Directory (ONSPD) and the DfE to control for local density

by calculating the number of premises in each postcode (which is fairly stable over time) and the number of students (of all ages) per premise.

2.2.3. Telecom infrastructure and postcode broadband speed data

In order to construct measures of available speed, we use telecommunication network information, including the location of the universe of English LE stations (approximately 3,900) and their assignments to each of roughly 1.45 million full postcodes, which we obtained from OfCom, the UK telecom regulator. This data allows us to link each residential address to its connected telephone LE and to allocate each household to a unique LE catchment area. We exploit the richness of this dataset to devise the empirical strategy set out in Section 3.

In Section 5, we use additional data on postcode-level realized internet speed from Ofcom, the British telecom regulator in the UK, to estimate the distance-to-LE speed relationship.⁹ The major fixed-line broadband suppliers (ISPs) provide data on individual speed tests to Ofcom, which aggregates the information by area in different years. The data from these suppliers cover over 80% of the market.¹⁰ Data at the finest geographical level, the postcode, have been available since 2012 and are published yearly, which does not directly align with our test score data period of 2005–2008. Note that our approach to combine linked postcode-to-exchange station telecom network data with data on the local average internet speed experienced by households is not dissimilar to other papers such as Ahlfeldt et al. (2017) and Amaral-Garcia et al. (2019).

Average postcode-level speed is calculated from information on millions of active broadband connections provided to the regulator and is based on *modem sync speed*, which captures the highest possible speed at which data can be transferred across the line with the use of a particular DSL technology (OfCom, 2012). The indicator captures the speed at which the modem in a customer's home connects to the equipment in the telephone exchange, and it is directly related to the subscription package headline speed. This way of measuring the line speed contrasts with speed tests obtained using modems at home and performed by users, who usually report slower speeds, which are affected by the time of the day at which the data transfer is done, the number of devices connected simultaneously and the quality of home software and internet equipment. Postcode average speeds are a reflection of the subscriptions to different broadband packages, which results in a mix of technologies: primarily ADSL but also cable or fiber internet.

A potential concern is the deployment of non-distance-sensitive technologies in Britain from 2008 (cable) and 2010 (fiber). Ofcom information for small geographies was only made accessible from 2012, when superfast technologies were already available in some areas. Importantly, even if in 2012 68% of England already had access to superfast broadband (yielding speeds over 30 Mbit/s), approximately 75% of the subscribed broadband connections were still using ADSL technology. In this sense, local averages for 2012 are the result for a majority of ADSL connections, which are sensitive to distance to the LE. Furthermore, since most packages offered 8–10 Mbit/s headline speeds (OfCom, 2009b) in the period 2005–2008, in our estimations we focus on postcodes with average (download) speeds that are realistic for this sample period.

⁹ These data are available from the Ofcom Infrastructure reports – now called Connected Nations – accessible via the Ofcom webpage and the National Archives webpages.

¹⁰ The suppliers include BT, Virgin Media, Everything Everywhere, O2, KCom, TalkTalk and Sky.

2.2.4. Survey microdata on broadband take-up and computer use at home

To further investigate the mechanisms behind the effect of broadband speed on educational achievement, we use data from the British Household Panel Survey (BHPS) in 2007. The BHPS is a large longitudinal survey representative of the UK population, with each wave covering about 15,000 geo-referenced households. We used the survey version with home location small geography identifiers, Lower Layer Super Output Area (LSOA), which we used to link home residents' location with respect to the telecom infrastructure. These are small neighborhoods, comprising between 400 and 1,200 households and have a usual resident population between 1,000 and 3,000 persons.

We combine the individual and household respondents' questionnaires and focus on wave Q (year 2007), which is the only one that includes questions related to the use of computers at home and the availability and use of the internet. First, we explore household-level questions regarding the availability of computers, internet connections and broadband subscriptions at home. Second, we examine information regarding computer use at home for individuals 15 to 65 years of age. Individuals have to answer whether they use the computer at home for different tasks, such as playing games or doing educational work. In addition, we use household background characteristics such as socio-economic status group, gender, household composition, housing tenure and age in the empirical estimations.

2.3. Summary statistics

Table 1 provides the descriptive statistics of key variables in the whole and our baseline estimation sample, with mean values and standard deviations. In the full sample, our data cover slightly more than 1.1 million students living in over 400,000 postcodes and attending more than 2,860 schools in England over the period 2005–2008.¹¹ As we discuss below, our estimation sample is constructed by focusing on households within 300 m of an LE catchment area boundary segment.

Panel A provides the descriptive statistics of our outcome variables for the subject-specific tests and the mean of the three. Panel B of Table 1 provides summary statistics on these two distances for both the full and estimation samples. We explain below how these measures are constructed. Panel C reports the pupil-level characteristics and shows that the vast majority of the pupils are white, approximately 14% are entitled to free school meals, and students live on average 2.5 kilometers from their schools. Panel D displays the postcode-level characteristics, which show similar values in the proportion of white and FSME students at the local level, a majority of community-type schools and an average house price of approximately £190,000. The table shows that the composition of pupils and area characteristics for the whole sample is highly similar to those of the estimation sample.

3. Empirical strategy

3.1. Sorting issues and identifying variation

To estimate the effects of home broadband on education outcomes, a major identification challenge has to be addressed: household-level observed broadband speed – i.e., package choices – is likely related to learning outcomes through confounding factors that are difficult to directly control for. We refer to this as *active* sorting; e.g., better-off households invest in better connections to boost outcomes. As a result of this type of sorting, using data on observed broadband speeds is

¹¹ The raw data include approximately 500,000 pupil observations per year. To prepare the sample for our empirical strategy, we exclude observations that (i) have implausible or inconsistent values, (ii) are assigned to segments that only have observations on one side, thus making it impossible to perform within-segment-year comparisons, (iii) cannot be linked to school or local area characteristics, or (iv) are located in postcodes that had broadband services enabled for less than six months.

problematic. The approach used by the existing literature is therefore to focus on variation in *available* broadband speeds, which depends on location choices but not on broadband subscription choices.

In this context, we exploit a feature of the DSL-broadband technology: a salient feature of the technology is that once a home is connected to a broadband-enabled LE station, the *available* connection speed depends on the length of the copper wire connection between the residence and the LE station.¹² We use this characteristic by exploiting “jumps” in distance to the LE across catchment area boundaries.

In the UK, the location of LE stations was determined during the deployment of the English landline telephony network, which mainly occurred before and during World War II. Importantly, distance to the LE station did not affect the quality of traditional telephone services, so the setup of the network was designed to maximize the number of connections from the minimum number of stations (with the goal of achieving cost-effectiveness). However, while we believe that it is unlikely that households in the past or present actively sort on the basis of distance to the connected exchange station, there are several reasons to believe that LE station location is far from random and is potentially correlated with other local neighborhood characteristics that do matter for household sorting. One can argue that most households are not likely to be aware that the speed of their broadband connection is related to their location choices, and even if they are, they might not precisely know where LE stations are located. Households might not be located at different distances from the LE because they know and care about the speed-distance relationship (abstracting from *active* sorting). However, they might sort with respect to other geographical features that are also correlated with station location. For example, LE stations appear to have been placed at central locations (local town centers) that were also close to major road junctions for hosting the exchange switchboard infrastructure. We refer to this as *passive* sorting. As local geography correlates with both the location of LE stations and the location of households across space, comparing households located within an LE would lead to biased estimates.

To overcome identification issues related to *passive* sorting, instead of comparing different locations within LEs (e.g., households connected to the same LE but at different distances from the station), we use a strategy that compares variation in DSL-cable length across neighboring locations. This strategy is somewhat similar to Black (1999), which compares house prices across school catchment area boundaries. As we show later, house prices are fully balanced in our setting because these boundaries were unknown to the general population over the study period. To ensure that other local characteristics do not affect our estimates, we compare households located very close to each other and thus with similar geographical features but with connections to different LEs and hence with different broadband *availabilities*. These boundaries give rise to substantial cross-sectional variation in the quality of the available DSL-broadband speed due to discontinuous jumps in the length of the copper wire that connects residences on either side of the boundary to their assigned LE stations. This holds across any given boundary segment. In a nutshell, we compare the distance to the connected exchange of postcodes on both sides. We call the resulting discontinuous changes in average distances to the LE on each side of small boundary segments “jumps”.

Note that in this paper, we do not compare outcomes between DSL-connected and unconnected places but exploit differences in broadband speed quality across locations.¹³ The period for which we could

compare locations with internet dial-up connections to those already connected to a broadband-enabled LE is that encompassing the rollout of the DSL infrastructure, e.g., mostly 2000–2004. The technological upgrade between these two technologies would have allowed us to exploit a 10x increase in expected internet speeds; however, the take-up rate of broadband services before 2005 was negligible, reducing the probabilities of finding an identifiable treatment. Instead, we use data for the years 2005–2008, when broadband infrastructure was almost universally enabled and when take-up rates were already substantial and growing, and focus on differences on available internet speed across neighboring locations.

We argue and provide evidence below that using the variation in available broadband speed across local boundaries addresses the discussed endogeneity concerns. This variation has to be exploited at very small scales to avoid spatial confounders correlated with location and outcomes. To leverage the richness of the data, it is essential to use very disaggregated information, in terms of both geographical scale and sample size; thus, the available geolocated administrative data are key for the application of a robust empirical strategy. Next, we explain how we take this setting to our data and our estimation approach.

3.2. Construction of the discontinuity and treatment variables

The core of our empirical strategy is the construction of the boundaries of LE catchment areas. The first thing to note is that these boundaries do not coincide with any other administrative boundaries, in particular, school district boundaries. For this, we use information on the precise geolocation of the universe of English LE stations (approximately 3,900). In particular, we use the assignment of each of the English postcodes to the LE that provides telephone and internet services to its premises. There are approximately 1.45 million full postcodes in England. Each postcode contains approximately 15 households on average, and the postcode areas are often as small as a single building, especially in denser areas. Panel A of Fig. 1 illustrates the location of an exchange station A and the associated catchment area. Fig. A.1 depicts the LE catchment areas for all of England.

We match individual postcodes to their closest exchange station boundary segment. First, we extract segments that are unique borders (lines) between two different LE stations from the boundary for each catchment area. This can be done by cutting the polygon boundaries at intersections. Second, the boundaries are divided into smaller segments (henceforth called boundary segments), which are on average 3.2 kilometers long (S.D. of 1 kilometer). In the third step, we assign all postcodes in England to the particular boundary segment that it is closest to, conditional on which LE they are connected to.¹⁴ This determines which boundary segment each postcodes belong to. Panel B of Fig. 1 shows such a particular boundary segment in red and close postcodes matched to it as little dots. Note that to compare households with similar geographical surroundings, we only use postcodes within 300 m of the boundary segment in our preferred estimation sample. Robustness checks on the sensitivity of our findings to the baseline sample selection are discussed extensively in Section 4.4.

For the boundary-postcodes, e.g. those assigned to a particular side of a boundary-segment, we can now compute two distance measures that are required for the construction of the treatment and SRD variables. First, we calculate the Euclidean distance from the postcode centroid to the connected LE station; this approximates the connecting copper cable length, which is a measure of internet speed quality.¹⁵

¹² Distance to the LE is not the only driver of the variation across DSL subscriptions, as other factors can also affect observed speed, such as the quality of the hardware and software used, the number of simultaneous users in the household, the day of the week and time of day, the size and upkeep of the LE station and other technical factors such as varying quality of in-house wiring, unconnected microfilters, or varying performance by the ISP. See OfCom (2009a, 2009b) for more details. See Ahlfeldt et al. (2017) for estimates of the determinants of household observed broadband speed.

¹³ Additional technological and regulatory changes also took place during our period of analysis; however, by comparing within-year cross-boundary segments, we focus solely on cross-sectional variation to obtain our estimates.

¹⁴ We know the precise geolocation of the postcode centroids using the British National Grid Eastings and Northings to the 1 meter precision from the National Statistics Postcode Directory.

Second, we calculate the distance between the postcode centroids and their boundary segments. This will allow us to compare postcodes that are close to each other.

It is important to reiterate that there are two different types of distances. The first is the distance from the postcodes to the connected LE station, which is an important determinant of the available broadband speed. This distance increases as we approach the boundary segment and changes discontinuously when we cross an LE boundary segment. This is the distance measure that gives rise to the variation in broadband speed across boundary segments. The second measure is the distance to the LE boundary segment. This distance is used to identify close neighbors, i.e., to select which locations we use as comparisons within a short segment, and it is our SRD running variable.

On each side of a boundary-segment, postcodes differ individually in their copper cable lengths, but on average, the group on each side is relatively closer or further away from its connected LE station. To make this geographical setting operational, we calculate the *average* distance to the connected LE of the postcodes on each side of a boundary segment. The side that has shorter average distances is labeled as the *Fast* side, and the other side is defined as the *Slow* side. The difference in the average distances between the two sides captures the “jump” in average cable length when we cross the boundary segment; from it we identify the impact of changes in speed broadband.¹⁵ This is illustrated in Panel B of Fig. 1. Here, the average exchange distance for the boundary postcodes in catchment area A is larger compared to catchment area D. As a result, the postcodes on the boundary segment connected to D are labeled the *Fast* side, whereas the other side is the *Slow* side.

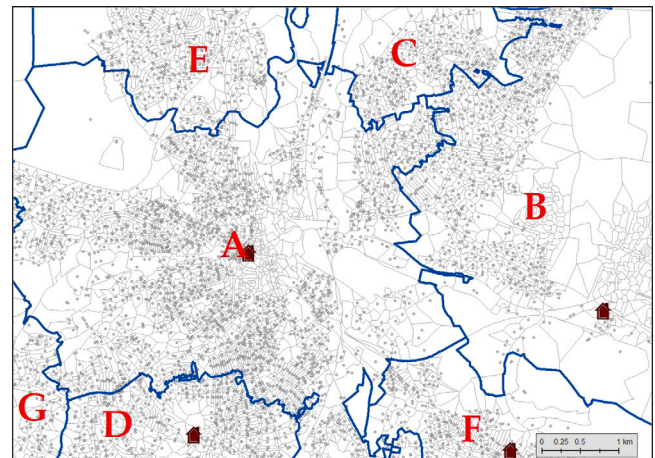
In the previous example, all the postcodes in the *Fast* side are on average but also individually closer to their connected LE station than all the postcodes in the *Slow* side (sharp RDD). However, it could be the case that two addresses located on different sides of the segment are not necessarily different in the way that we would expect. Due to the irregular geographic shape of the boundaries, some households with shorter cables (longer cables) might live on the on-average slower side (faster side). If the jump in average distances between sides is attenuated because some postcodes are “assigned to the wrong side”, this could result in attenuation bias. We illustrate this situation in Panel C of Fig. 1.¹⁷ To take into account these cases, we implement a fuzzy spatial RDD using an IV strategy, which we explain in detail in the next subsection.¹⁸

¹⁵ A shortcoming of our approach (common to other, similar papers) is that we can only calculate crow-fly distances between the centroids of the postcodes of the location of pupils' homes and the LE station to which they are connected; in urban areas, there can be a substantial gap between this and the actual length of the connection cable (OfCom, 2009b). Nevertheless, given the very small scale of our geographical units of observation, we can approximate this in a more precise way than other studies that use data for larger geographical units.

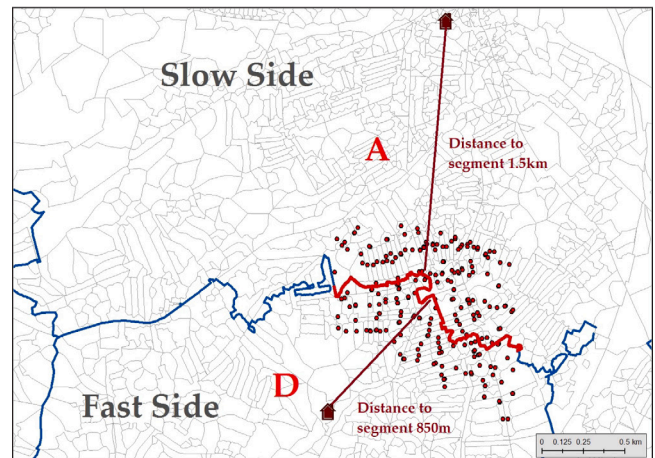
¹⁶ Sometimes the jump between the two sides is small, if the variation in average distances is relatively low. For this reason, we exclude segments in which the jump is below 100 m, and for our main results, we focus on segments with jumps of at least 300 m. We extensively discuss the robustness of this choice in Section 4.4.

¹⁷ For segment AB, the average distance to the connected LE is quite similar on both sides, approximately 1.6–1.7 kilometers. The shape of the segment is irregular and slightly diagonal, tilted to the west. Two sets of postcodes are selected to explain this situation. The triangular-shaped ones are both around 3 kilometers away from the LE station, but the shorter of the two (3,043 m) is located on the *Slow* side, and the longer segment in the pair (3,049 m) on the *Fast* side. The pentagon-shaped postcodes display the inverse situation: the one with the longer distance, 2,467 m, is located on the *Fast* side, while that with the shorter distance (2,402 m) is on the side with longer distances on average.

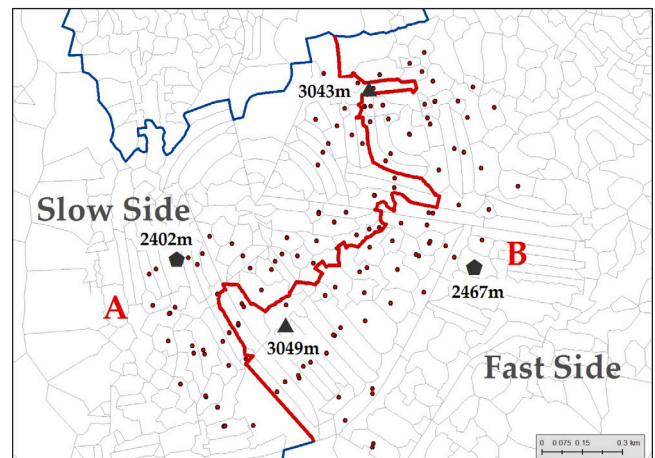
¹⁸ For completeness, we also present (attenuated) sharp SRD estimates, see Table A.3.



(a) Overview of LE catchment areas and boundary segments



(b) Postcode sample boundary A–D segment



(c) Postcode sample boundary A–B segment – Fuzzy

Fig. 1. Graphical illustration of empirical strategy.

Notes: The blue lines represent the boundaries of the LE catchment areas. Within them, light grey lines show the postcode borders, and the light gray dots represent the precise location of the postcode centroids at the 1 meter-resolution precision. The building symbols display the exact locations of LE stations within the LE catchment areas. Subfigure (a) shows an overview of several LE station catchment areas, with underlying postcode polygons and centroids. One letter is allocated to each LE. Note the irregular shape of the borders and the fact that LE stations are not always located in the center of the LE catchment area. Subfigure (b) and (c) zooms into two specific LE catchment area boundaries: A–D (b) and A–B (c), in which we have highlighted a particular segment in red. The red dots mark the postcodes located within 300 meters of the red boundary segments. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.3. Specification and IV strategy

Our goal is to estimate the causal effect of broadband internet speed on test scores for 14-year-old students. The basic framework of our analysis, capturing the relationship of interest, is the following:

$$TestScore_{ipnlst} = \beta ProxLE_{pl} + g(D_{pn}) + X'_i \Lambda + Z'_{is} \Theta + A'_{pt} \Phi + L'_p \Psi + \delta_{nt} + \epsilon_{ipnlst} \quad (1)$$

where $TestScore_{ipnlst}$ is the percentile rank in the KS3 test of pupil i living in postcode p in boundary-segment neighborhood n associated with LE station l and attending school s at time t ; $ProxLE_{pl}$ is the proximity (negative distance) to connected LE station l from home postcode p ; X'_i is a vector of student background characteristics, such as preinternet student performance on the KS1 test, gender and free school-meal eligibility status; Z'_{is} is a vector of characteristics of the school attended by pupil i , such as school type and distance between home and school; A'_{pt} is a vector of postcode-year-specific characteristics, such as local average housing prices, share of students eligible for free school meals, and white (population) pupils; L'_p is a vector of time-invariant postcode attributes such as density (e.g., number of delivery points) and distance to different amenities (e.g., nearest rail station or road); δ_{nt} is boundary segment-by-year fixed effects, which guarantee that we are comparing students within the same segment-year of the LE boundary; and ϵ_{ipnlst} is the error term. To ease interpretation of the estimates, we measure $ProxLE_{pl}$ as “negative distance”, e.g., proximity to the LE station. Thus, β captures the changes in test scores when we come closer to the LE by one meter and available broadband speed improves. We cluster the standard errors at the segment-by-year level.¹⁹

To apply the empirical model to the data, we need to specify two additional pieces of information. First, the definition of $g(D_{pn})$ captures the relationship of the postcode distance to the boundary segment. This deterministic function has the spirit of the running variable in a nonspatial regression discontinuity design. In our setting, we are interested in controlling for distance to the boundary segment because of *passive* sorting, as explained in Section 3.1 above. As we have a large sample of pupils, even when we restrict observations to postcodes within 300 m of the border, we can carefully control for distance to the boundary segment, where each side is connected to a different LE station. This means that we can effectively compare the test scores of students with differences in home broadband access who live very close to each other.

Our preferred control function for the relation with the running variable is the following:

$$g(D_{pn}) = \sum_{b=1}^{B_s} (\gamma_b^{Slow} Slow_{pn} * I(DistSegment_{pn} = b) * DistSegment_{pn}) + \sum_{b=1}^{B_f} (\gamma_b^{Fast} Fast_{pn} * I(DistSegment_{pn} = b) * DistSegment_{pn}) \quad (2)$$

where we control for the distance to the LE invisible boundary segment by using distance bin dummies of 100 m (b) interacted with the distance to the boundary segment on both sides of the cutoff. In terms of the notation, $Slow_{pn} * I(DistSegment_{pn} = b)$ is an indicator variable for postcode p on the *Slow* side of boundary segment n that equals one if the distance of postcode p to segment n is within bin b . $Fast_{pn} * I(DistSegment_{pn} = b)$ is defined analogously for all postcodes on the *Fast* side of the boundary segment n . This semiparametric approach allows for more flexibility in controlling for distance to the segment. Instead of imposing a certain functional form on the polynomial, we estimate the coefficients γ_b^{Slow} and γ_b^{Fast} for each small distance bin for the segments (i.e., 0–100, 100–200, and 200–300), thereby capturing the shape of the polynomial in a flexible way on each side of the

boundary segment. This approach is flexible and avoids either oversimplifying the underlying relationship (as would be the case if, for example, we used a linear polynomial) or over-fitting by using high-order polynomials (Gelman & Imbens, 2019). The results are robust to alternative definitions of distance bins, to the use of distance as a continuous variable, and for a more flexible approach of higher-order polynomials.

One final step is required for estimation. In order to overcome the attenuation bias induced by the assignment of postcodes into *Fast* (closer) and *Slow* (further away) sides, we use an instrumental variables strategy and a fuzzy SRD specification. Using the information on which side of each boundary segment has a lower average distance to the LE, we construct segment-side-specific *Fast* dummies that we use as instruments for the actual postcode-level distance to the LE. The first-stage equation is:

$$ProxLE_{pl} = \pi Fast_{pn} + g(D_{pn}) + X'_i \Lambda + Z'_{is} \Theta + A'_{pt} \Phi + L'_p \Psi + \delta_{nt} + \epsilon_{ipnlst} \quad (3)$$

where $Fast_{pn}$ is a dummy variable equal to one if student i living in postcode p is located on the *Fast* side of LE station boundary segment n and zero otherwise. As above, $ProxLE_{pl}$ is measured as proximity to the connected LE station. The coefficient π captures how much closer postcodes are to their connected LE stations on the *Fast* side relative to the average distance on the *Slow* side.

Since the telephone LE station boundaries are historically given and under the assumption that households do not sort on each side of the boundary within the boundary segment, we can focus on households whose residences are located in the vicinity of the LE boundary segment, considering those on the *Fast* side very similar to those on the *Slow* side. The broadband speed assigned to these households can be considered “locally” randomly assigned within segment-by-year. Assuming that in the absence of the treatment, the outcome variable is a smooth function of distance to the LE boundary, the causal effect of broadband internet speed is identified by comparing outcomes for pupils who live close on the *Fast* side of an LE station boundary (treatment group) with those who are near but live on the *Slow* side (control group). This effect is captured by the IV estimate of β in specification (1). This strategy estimates a local average treatment effect (LATE) of broadband internet speed on student performance by comparing “lucky” households that are supplied with faster broadband access to otherwise similar counterparts that were “unlucky” in terms of being supplied with slower broadband access. A specific feature of this strategy is that it generates variation over multiple thousand telephone LE station boundary segments, which vary in distance to the LE station and in the jump in distance across the boundaries.

This identification strategy is related to those of Ahlfeldt et al. (2017), Falck et al. (2014) and Amaral-Garcia et al. (2019). Falck et al. (2014) estimate the effects of information disseminated over the internet on voting behavior in Germany between 2005–2008, where entire (small) locations happened to be located too far away from an exchange station to access broadband. In particular, towns farther than five kilometers from an exchange could not obtain any broadband internet without costly further technological upgrades. This characteristic allows the authors to exploit differences in outcomes between places that were connected to the DSL network and those that were not. However, this approach is not directly applicable to the British context because of a much denser network of LE stations, related to the smaller size of the country, with all places connected to the network, and the relatively quick rollout of the broadband infrastructure. Similarly to us, Ahlfeldt et al. (2017) and Amaral-Garcia et al. (2019) exploit discontinuities in the proximity to connected stations across LE catchment areas boundaries. They exploit changes in local speeds induced by technological upgrades over time. In contrast, we exploit spatial cross-section variation in distance to the connected LE station within a year for a stable technology (ADSL).

¹⁹ Table A.7 shows that different clustering choices for our preferred specification do not change our conclusions.

3.4. Estimating the impact of broadband speed on test scores

In a simple framework, faster broadband internet could affect teenagers' school performance in different ways. High internet speed allows students to access more online content per unit of time. If test scores are determined in a learning production function, we can think of speed impacting learning productivity. For each hour of study, students can access more information, shifting the learning production function upwards if, for example, they can access more learning resources such as Wikipedia or online interactive materials. However, broadband could affect learning investment by reducing study hours if students divert time to non-learning online activities, such as gaming or using social media. This second channel could be more or less relevant depending on whether online distractions replace offline distractions. The first channel would have a positive effect on human capital formation, while the second would have a negative effect. Coefficient β in Eq. (1) captures the net impact of these channels, and our aim is to identify an unbiased estimate of this coefficient.

Henceforth, the causal estimation of β informs us about the impact of *available* broadband speed on test scores. It captures how much test scores change as we approach an LE station via the relationship between copper cable length and potential speeds. We estimate a LATE impact by using a proxy of broadband speed that changes over space. Our treatment measures local availability of faster broadband rather than student-specific faster connections and provides a measure of the intent-to-treat (ITT) impact.²⁰ While the estimation of this parameter is desirable from an identification point of view, it is also a relevant policy variable. By investing in network improvements and expansions, public policy can influence the availability of fast broadband in different locations, while afterwards, particular households might sort into different broadband packages and use the internet in different ways for reasons that might correlate with the outcomes of interest.

However, we are ultimately interested in the impact that broadband speed changes have on student performance. For the reasons explained above and due to the lack of data appropriate to the time period and geographical scale of our analysis, we cannot directly estimate this impact in the data. Our approach is to employ additional data to compute the distance-speed relation present during our study period, combining linked postcode-to-exchange station telecom network data with data on the local average internet speed experienced by households.²¹ We compute the Wald estimate by dividing our LATE coefficient (effect of proximity to LE station on test scores) by this parameter (proximity to LE station on average speed) to obtain a measure of the impact of changes in speed on test scores, and calculate the standard error using the Delta method.

In a last step, similarly to [Detting et al. \(2018\)](#) and [Zuo \(2021\)](#), we re-scale our estimates to provide a measure of treatment-on-treated (TOT) effect. We adjust our Wald estimate by the relevant population of interest. We do this using additional survey data to obtain information

on the appropriate broadband take-up rates during the period of analysis (those of families with teenagers during 2005–08 in England) and on the proportion of pupils using the internet for school work. This allows us to assess the economic relevance of our results and contextualize the size of our estimates. We discuss our quantification estimates in Section 5.

3.5. Estimation of the mechanisms

This subsection uses the telecom network data matched to the BHPS to analyze the channels behind the effect of broadband speed on educational achievement. We perform two exercises: first, we test if household internet-related variables are balanced across fast-slow sides of boundaries, and second, we explore different computer uses across different age groups. We adapt the empirical strategy to the smaller survey dataset and compare individuals on the fast and slow sides of LE catchment area boundaries (reduced-form).

To test if broadband take-up and computer presence at home differ across the fast-slow sides of the boundaries, we ran the following specification:

$$Y_{ihmlt} = \beta Fast_{hm} + g(D_{hm}) + X'_i \Omega + Z'_h \Lambda + \delta_{mt} + \epsilon_{ihmlt} \quad (4)$$

where Y_{ihmlt} is a dummy which indicates if household i living in LSOA h in boundary neighborhood m associated with LE station l at time t has internet/broadband/computer at home; $Fast_{hm}$ is a dummy variable equal to one if household i living in LSOA h is located on the *Fast* side of LE station boundary m and zero otherwise; $X'_i \Omega$ includes head of household and general household characteristics including age, gender, number of children, housing tenure and socio-economic status; $Z'_h \Lambda$ is a vector of LSOA socio-economics area characteristics; δ_{mt} are boundary-by-year fixed effects, which guarantee that we are comparing households within the same LE boundary-year; and ϵ_{ihmlt} is the error term. We cluster the standard errors at the boundary-by-year level. The relationship of the LSOA distance to the LE boundary is captured by the control function $g(D_{hm})$, which includes distance bin dummies of 100 m interacted with distance to the boundary, differently estimated on each side.

To dig deeper into the mechanisms driving our findings, we then explore a set of computer uses (for work, for education purposes, to play games) by individuals of different ages. For households that have broadband connections at home and conditional on using the computer to access the internet, we estimate the following specification:

$$Y_{ihmlt} = \beta (Fast_{hm} * AgeGr_{ihmlt}) + \gamma AgeGr_{ihmlt} + \rho Fast_{hm} + g(D_{hm}) + X'_i \Omega + Z'_h \Lambda + \delta_{mt} + \epsilon_{ihmlt} \quad (5)$$

where $AgeGr_{ihmlt}$ identifies individuals i aged in different age groups in LSOA h in boundary neighborhood m associated with LE station l at time t . We test the interactions for two age groups: teenagers (aged 15–17) and young adults (aged 18–24). This specification is akin to a difference in discontinuities design. In each estimation, we compare two age groups (for example, teenagers vs adults) across boundaries (fast and slow sides). This setup allows us to explore whether teenagers use computers differently from adults in the same household, and whether teenagers on the fast side behave differently from those on the slow side. Hence, we can focus on the population of interest, i.e. teenagers. While γ captures the differential use for teenagers, the interaction term, β , captures the differential computer use of teenagers on the fast side with respect to those on the slow side.

For consistency with Section 3.2, we focus on boundaries with jumps of at least 300 m. Our baseline regression focuses on individuals living within one kilometer of the LE catchment area boundary to improve statistical power and compare across boundaries instead of boundary-segments. In the regression, we include the same area controls as in the main test scores estimates, as well as additional individual and household controls.

²⁰ Related papers also provide ITT estimates ([Ahlfeldt et al., 2017](#); [Amaral-Garcia et al., 2019](#); [Detting et al., 2018](#); [Geraci et al., 2022](#)).

²¹ In [Ahlfeldt et al. \(2017\)](#), the authors estimate a hedonic price model of the effect of broadband speed on individual house prices using a prediction of local yearly broadband speed constructed from a separate dataset. To overcome the lack of availability of postcode-year speed data, the authors predict postcode-level speeds served by different LE with different technologies for years 1995–2010 using a large individual broadband speed test dataset in years 2009–2010, which includes rich information on internet packages, type of technology and distance to LE. [Amaral-Garcia et al. \(2019\)](#) rely on their estimates to construct speed predictions by ADSL technology and geographical aggregation (LSOA). They use the predicted speeds to assess the impact of broadband internet speed on C-sections. These strategies are close to a two-step procedure, similar to the one we implement by means of a Wald estimate (with adjusted S.E.).

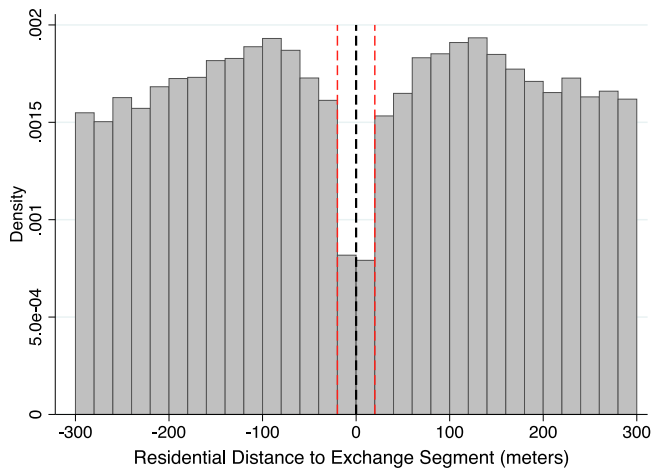


Fig. 2. Distribution of pupil distance from residential distance to exchange segment. *Notes:* The figure shows the density of pupils in our preferred estimation sample at the LE boundary segment. The black dashed vertical line presents the LE station boundary segment. The red dashed vertical lines show the boundaries of the donut strategy. Each bar contains bins of 20 m distance.

4. Results

4.1. Internal validity of the empirical strategy

The internal validity of the fuzzy SRD requires that there is no endogenous sorting on either side of the LE boundary segment. In the case of this study, there are two key features that make manipulation at the LE station boundary segment highly unlikely. First, LE catchment areas are either invisible or unlikely to be known by households. Second, LE boundaries do not coincide with administrative boundaries of any kind, such as school district boundaries. The fact that both our identification strategy is not bound to other administrative boundaries where variables affecting our outcome of interest might change discontinuously at the cutoffs, and that we have granular data to use a data-driven approach to generate boundary segments, provide a unique advantage to estimate the causal effects of home broadband speed.²²

Fig. 2 presents a histogram representation of pupil density at the LE station boundary segment for our preferred estimation sample. It allows for visual inspection of whether bunching takes place. It reveals no evidence of systematic manipulation of residential distance to the LE boundary segment around the thresholds. However, we observe that the density drops as we get closer to the border, particularly within 20 m of the LE boundary segment (indicated by the red vertical dashed lines). This drop is a byproduct of the data construction: postcode distances to the boundary segment are measured from the centroids, and there are very few oddly shaped postcodes where the centroid falls within 20 m of the boundary segment. The fact that the drop is of the same magnitude on both sides of the discontinuity is a positive indicator that it is unlikely that this feature has a differential effect on one particular side of the LE boundary segment. For robustness, in the estimations, we drop observations closer than 20 m.

We also perform more in-depth analyses and formal tests of bunching, as reported in Section 4.1. We formally test for bunching following McCrary (2008) and Cattaneo et al. (2018). The McCrary (2008) test fails to reject the null of no significant jump at the LE boundaries.²³ Cattaneo et al. (2018) propose a test that is robust to bandwidth

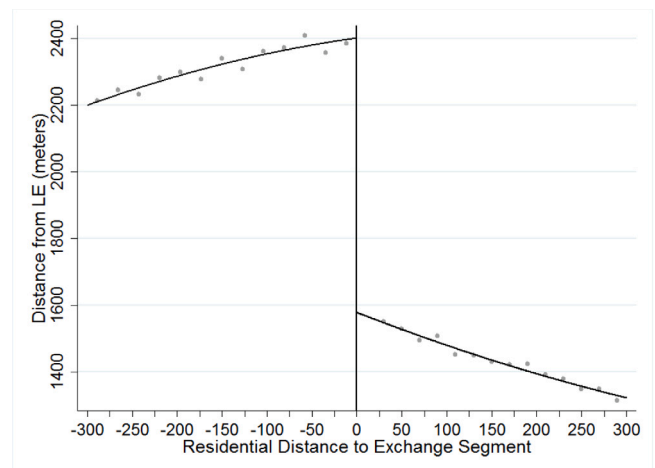


Fig. 3. Distance from the connected LE station jumps across the boundary segment. *Notes:* The dots represent the average distance to the LE station per 25-meter interval of residential distance to the LE boundary-segment boundaries. The solid lines are fitted values from a third-order polynomial approximation, which is estimated separately on both sides of the cutoffs. “Residential Distance to Exchange Segment (Meters)” refers to the residential distance to the LE boundary-segment boundaries. The black vertical line is the stacked LE station boundaries.

selection issues. We fail to reject the null hypothesis of no discontinuous jump at the LE boundary segment.

Since the density drop within 20 m is a byproduct of the geographical resolution of the data, we use the so-called donut strategy, which excludes observations within 20 m of the discontinuity in our main specifications, following Angrist et al. (2019) and Leuven and Løkken (2020), among others. Discarding these observations improves the precision of our estimates since we eliminate potentially rare postcodes and spillovers across LE boundaries.²⁴

Another direct check of a violation of instrument exogeneity is to test whether pupil, school and area/postcode baseline characteristics are “locally” balanced on either side of the LE boundary segment. If these variables are unbalanced on either side of the boundary segment, it would indicate selection problems around the discontinuities. Table 2 tests the balance of a battery of background characteristics. The first column of Table 2 regresses the main outcome on the predetermined background characteristics. The regressions further control for distance bins and include segment-by-year fixed effects. The results of this column show that the background characteristics are economically and statistically important in explaining the variation in the outcome variable. We strongly reject the null hypothesis that these variables are jointly equal to zero. The third column of Table 2 displays the results of local linear regressions for each of the predetermined pupil, school, density, area socioeconomic, housing price and amenity characteristics. We show that most of the regressors are not significantly different from zero, and the coefficients are very small. In particular, important determinants of our outcome variable, such as individual preinternet scores, free school meal eligibility and housing prices, are statistically indistinguishable from zero.²⁵

²³ We also test for various alternative residential distances to the LE boundaries. The results are robust to different specifications. See Table A.2.

²⁴ At the boundary segment, very local spillovers might exist where households on the *Slow* side could connect to the WiFi routers of neighbors on the *Fast* side. The fraction of pupils who are discarded represents less than 1% of our estimation sample.

²⁵ Ahlfeldt et al. (2017) finds statistically significant impacts of faster broadband internet on house prices. Note that such a result does not raise concerns about the potential bias of our baseline estimates. As we explain in

²² In the data construction, we removed boundary segments that intersect with natural boundaries (rivers). While it is now possible to determine (rough) boundary locations online, this was not possible during the period studied.

Table 2
Balance of baseline student, school and area/postcode characteristics.

	Outcome average percentile rank score (1)	Slow side baseline mean (2)	Instrument Fast side nonparametric estimate (3)
A. Student Characteristics			
White	−0.592** (0.244)	0.78	−0.014* (0.007)
Male	−2.430*** (0.133)	0.5	0.005 (0.010)
Free School Meal	−13.749*** (0.206)	0.16	0.003 (0.008)
Pre-KS3 Score (KS1)	0.819*** (0.002)	43.89	−0.153 (0.508)
B. School Characteristics			
Log Distance Home to School	3.699*** (0.112)	7.39	0.012 (0.017)
Log School Size	1.305*** (0.140)	6.88	0.007 (0.013)
Community School	−9.951*** (0.214)	0.62	0.003 (0.009)
C. Density & Area Socioeconomics			
Log Number of Delivery Points	−0.944*** (0.132)	3.34	0.013 (0.015)
Log Number of Pupils per Premise	−1.754*** (0.121)	−1.76	0.016 (0.016)
Share of White Pupils	1.416*** (0.389)	0.77	−0.008 (0.005)
Share of Free School Meal Pupils	−22.724*** (0.354)	0.17	−0.001 (0.006)
D. Housing Prices			
Log Average Housing Price	13.310*** (0.270)	12.04	0.014 (0.009)
E. Amenities			
Log Distance to Closest Road	0.045 (0.103)	5.44	−0.013 (0.022)
Log Distance to Closest School	1.416*** (0.134)	5.83	0.033** (0.016)
Log Distance to Closest Supermarket	1.156*** (0.191)	6.56	0.013 (0.012)
Log Distance to Closest Rail Station	0.118 (0.259)	7.22	0.020** (0.009)
Log Distance to Closest Water Body	−0.144 (0.136)	6.5	−0.002 (0.017)
Joint F-test	10,381		1.23
Joint <i>P</i> -value	0		0.23
No Segm-years	7,096		7,096
No Observations	183,892		183,892

Notes: Column (1) regresses the main outcome on the predetermined background characteristics. Column (2) shows the baseline mean on the *Slow* side for the predetermined background characteristics. Column (3) performs local linear regressions for each of the background characteristics. Each regression controls for distance bins and segment-by-year fixed effects. F-tests (and corresponding p-values) are for the joint significance of the variables reported in each column. Standard errors clustered at the segment-by-year level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

We find that postcodes on the *Fast* side of the LE boundary segment are more likely to be closer to a school and a rail station and less likely to be white. However, the magnitude of the differences in these

characteristics with those on the *Slow* side of the LE boundary segment is very small (0.6%, 0.3% and 1.8% with respect to the baseline mean, respectively). Given the number of variables tested, it is not surprising to find some small imbalances in individual variables. In line with this, we fail to reject the joint test in which all the coefficients are equal to zero at conventional levels of statistical significance. This means that we do not find evidence that third factors, some of which are extremely important in explaining later test scores, change discontinuously across our boundary segment.

Section 3, we implement a different empirical strategy. While they exploit changes in local speeds induced by technological upgrades over time, we utilize spatial cross-section variation in distance to the connected LE station within a year. Thus, the fact that we find house prices are balanced on either side of the discontinuity (see Panel D of Table 2) does not contradict the main findings of Ahlfeldt et al. (2017).

Table 3
Fuzzy SRD estimates: Impact of exchange proximity on student performance.

Dependent variable: National exam scores at age 14 (KS3)				
	(1)	(2)	(3)	(4)
Proximity to LE (negative distance)	0.00123** (0.00049)	0.00132*** (0.00048)	0.00131** (0.00048)	0.00122** (0.00047)
Student pre-KS3 Score (KS1)	No	Yes	Yes	Yes
Time-Invariant Area & Amenities	No	No	Yes	Yes
Time-Variant Area & House Prices	No	No	No	Yes
Observations	183,892	183,892	183,892	183,892

Notes: The table shows the fuzzy SRD nonparametric estimates for the effect of the home distance to the LE station on test scores. All specifications include non-parametric 100 m distance bins and segment-by-year fixed effects, as well as basic student controls. Columns report different specifications to which we add sets of controls additionally. Each coefficient comes from a separate regression, where the running variable is the residential proximity to the LE station boundary. We only include boundaries where the distance jumps by at least 300 meters. The average distance in exchange jumps by 815 meters when crossing the invisible boundary (First-stage coefficient). Standard errors clustered at the segment-by-year level are reported in parentheses. ** $p < 0.05$, *** $p < 0.01$.

4.2. Discontinuities in distance to the LE station

We now investigate the discontinuities across the boundary segment in the distances to the connected LE stations. Fig. 3 plots the distance to the connected LE station as a function of household distance to the LE boundary segment. The left side shows the average “stacked” distances to connected exchange stations of all boundary segments on the *Slow* side. As we move closer to the boundary segment, the distance to the exchange increases. Then, there is a clear discontinuity. On the *Fast* side, the average distance to the (different) connected exchange in turn decreases as we move farther away from the boundary segment and closer to the LE station.

4.3. Impact on student test scores

Table 3 shows the fuzzy SRD nonparametric estimates on the average student percentile rank score. Each column presents estimates from increasingly saturated specifications. Column 1 includes distance bins and boundary-segment-by-year fixed effects required for the SRD, along with basic student demographics. Column 2 adds student test scores from a pre-broadband period in England (Key Stage 1 test scores from ages 6/7) and school-level controls. This inclusion effectively modifies the empirical strategy to a fuzzy SRD *value-added* design. Using this specification, moving one meter closer to the LE station increases the national KS3 exam performance at age 14 by 0.00132 percentile ranks, with results statistically significant at the 5% level.

Columns 3 and 4 introduce additional time-variant and time-invariant area and amenity controls from Table 2. None of these adjustments significantly affect the estimates, indicating strong balancing across the invisible boundaries. We consider the estimates from column 4 as our preferred specification, and we discuss more results from this below.²⁶

Economically, how significant are these effects? The estimates in Table 3 reflect the impact of moving one meter closer to the exchange on student national percentile scores. In Section 5, using additional data on speed, we provide direct estimates that quantify this positive effect in terms of broadband speed and usage, demonstrating its economic significance.

4.4. Robustness checks

²⁶ As explained above, due to the irregular geographic shape of several boundaries, it is essential to use a fuzzy SRD design since sharp RD estimates would suffer from attenuation bias. Table A.3 illustrates the magnitude of this attenuation bias: the sharp SRD estimates are approximately five times lower than the preferred fuzzy SRD estimates. The discontinued working paper by Faber et al. (2015) presents such sharp estimates without the Key Stage 1 controls, resulting in small and statistically insignificant estimates.

4.4.1. Sensitivity to sampling choices

All results discussed so far are estimated using the sample of postcodes within 300 m of the boundary segment, where the distance jump to the connected exchange across the boundary is at least 300 m. To clarify that these choices do not affect our conclusions, Table A.4 presents our preferred specification estimated with varying samples, from observations within 100 m of the boundary segment to those within 1 kilometer on either side. The effect sizes are slightly larger in samples closer to the boundary segment (0.00157 in the 100-meter sample) but are not significantly different from each other at conventional levels.²⁷

Table A.5 again utilizes our main sample of postcodes within 300 m of the boundary segment but restricts focus to boundaries based on the magnitude of the distance jump upon crossing to the *Fast* side. In the sample that includes all boundaries except those with distance jumps of less than 100 m, the average distance change when crossing to the *Fast* side is 664 m (see column 1, Panel A). In contrast, when excluding boundaries with jumps of less than 500 m, the average treatment effect intensifies to nearly one kilometer (column 5). Estimates of the per-meter effect of distance are smaller in samples that include more boundaries contributing little to the variation, i.e., in columns 1 and 2. Conversely, the per-meter effect remains reasonably constant in samples excluding (almost irrelevant) boundaries with jumps of 300 m or more. The trade-off between sample size and excluding boundaries offering limited variation motivates the choice of a 300-meter threshold as the baseline. Moreover, including many boundaries with small variations “on average” introduces more misclassification of local postcodes into the *Fast* and *Slow* categories.

4.4.2. Sensitivity to school factors

Table A.6 compares our baseline estimates (column 1) with four additional specifications that control for available broadband technology and other factors at the school level in various ways. We find that student test scores are higher when schools are closer to an LE station, indicating access to faster broadband, and as the number of months since ADSL upgrade increases, measured using an indicator variable for early adoption (column 2).²⁸

It is important to note that these school-level effects are not estimated using a causal strategy and should be interpreted cautiously. School proximity to the local exchange, which implies faster broadband availability, is associated with positive effects on test scores amounting to about one-third to one-half the size of the home broadband effect.

²⁷ Note that the results remain unchanged when we exclude area and amenities controls in observations very close to the boundary, such as those at 100 or 150 m.

²⁸ Fig. A.2 shows the distribution of the update timing for school exchange stations in 2005. The early indicator equals one for updates before 2001, zero otherwise.

The indicator for early adoption is also positive and significant, though it is important to remember that this estimate is not based on an SRD and is thus susceptible to *passive* sorting. However, as the central takeaway, our main SRD result for the effects of home broadband remains robust even with the inclusion of this school-level control.

In columns 3–5, we incorporate various fixed effects into our specification. Column 3 includes school LE fixed effects that absorb the early adopter variation. Column 4 further adds school fixed effects, and column 5 introduces school-by-year fixed effects.²⁹ These rigorous specifications are feasible because the secondary education system in England does not operate with catchment areas, allowing a sufficient number of students from both sides of each boundary segment to attend the same or different secondary schools. These highly saturated models demand substantial computing power. From this specification, we estimate an effect of crossing the boundary of 0.00092 per meter. While the point estimate is slightly lower than previous estimates, it is not statistically different, and the coefficient remains significantly different from zero at the 5% level. One reason these estimates of the effect of home broadband on test scores might be smaller is that school fixed effects can partly absorb interactions between home and school-level technology, which we explore and confirm in the next section.

4.5. Heterogeneous effects of home broadband

4.5.1. By school technology

We know from the reviewed literature and our suggestive analysis in Section 4.4.2 that school technology can matter for learning. We now examine heterogeneity in the home broadband effect by splitting the sample along the two school-level ICT variables that we have data on, and which showed significance for test scores in Table A.6: the school distance to the connected exchange station and an indicator for early broadband upgrades. Columns 1–4 of Table 4 show estimates for quantiles in school-exchange proximity (from closer – faster – to further away): the effect is strong and positive only in the first quantile, e.g. those schools which have faster potential speeds. This means that the positive impacts of home broadband emerge only if schools have access to faster broadband. If schools are not well connected, the effect of home broadband is precisely zero (columns 2–4). Columns 5 and 6 mirror this: significant positive impacts of home broadband on test scores can only be estimated for schools with local exchanges that received earlier ADSL upgrades, where it is more likely that the penetration of fast broadband is higher and the usage more widespread. We discuss the implications of this heterogeneity in more detail when we conclude.

4.5.2. By student background characteristics

We analyze heterogeneity along the following student predetermined characteristics: student gender, prebroadband test scores, ethnicity and free school meal eligibility (See Table 5). Some interesting patterns emerge. First, the effect is not driven by only one gender. The effect per meter of distance is both more positive and more significant for girls than for boys (0.00125 vs. 0.00098, respectively). However, both estimates are positive. Second, the positive effects are strongest for high achievers based on a median split of the prebroadband KS1 test scores. The effect for high achievers is 0.00156 percentiles per exchange proximity meter (column 4) and 0.00096 for low achievers

(column 5). Third, nonwhite pupils have a larger estimate (0.00211 per meter, column 7), but this effect is also less precisely estimated than that for white pupils (0.00114, column 6). Last but not least, only students who are ineligible for free school meals benefit significantly, with an estimated effect of 0.00132 percentiles per meter in comparison to 0.00073 per meter (approximately half the size) for students with free school meal eligibility.

Noting the differential impact by FSME status and ability, in Table A.9 we examine this further. We estimate the impact by ability-FSME subgroups and find that the effect is driven by students from higher SES backgrounds, e.g. those not eligible for free lunches. The results of columns (1) and (2) show that the impact is driven by students who are non-FSME. The effect is stronger and more precise for those who are in the higher ability (KS1 score) group. For pupils in the FSME group, even if the sample is smaller, we estimate very precise zero impacts. Even if, due to a lack of data, we cannot estimate the precise mechanisms behind these results, we can posit two potential reasons. The first one is parental involvement: parents with higher SES are more likely to be more knowledgeable about the potential educational benefits of faster broadband, and supervise or help their kids to use the internet productively. Secondly, in richer households, even though there could be more devices connected at the same time, realized speeds could be higher as these households purchase better internet packages, for the same potential speeds.

These resulting coefficients are not distinct from each other at conventional levels of statistical significance. However, the overall pattern is of interest, with the most positive effects for girls, high achievers and students who are not eligible for free school meals. This result is consistent with that of Dettling et al. (2018), who find that the impact of high-speed internet on college applications is concentrated among white students with more educated parents and mainly located in urban and high-income areas. Moreover, Malamud et al. (2019) finds no significant effects of home internet access on student achievement for students enrolled in low-achieving primary schools in Peru. Potentially, the smaller effect on the below-median prebroadband KS1 test score and free school meal-eligible groups may partly reflect the trouble that struggling students have in developing effective study strategies for learning (Angrist & Lavy, 2009; Fryer, 2011), in our case in a home online environment. While none of the groups have negative effect estimates, these results still speak to the hypothesis that home broadband access might exaggerate existing educational inequality by achievement and family affluence.³⁰

Last but not least, we also analyze heterogeneity across subjects, not finding much as the confidence intervals of the estimated coefficients are overlapping. This additional analysis is presented in Table A.8.

4.6. Effects in the longer-run: age-16 KS4 scores

We exploit our student register data to examine whether the effects are lasting or fading out in the medium run. For this, we consider the national age-16 Key Stage exams that our cohorts are taking in the years 2007 to 2010. Notably, this period is now extending into post-2008, when alternative technologies became available, potentially diluting the treatment. There are, in particular, smartphones and fiber cable connections that do not show the distance-speed relations of the ADSL technology.

Note that in order to compute the age-16 student performance variables we had to categorize exams into the three subjects English,

²⁹ In further analysis, we also added by-subject fixed effects, i.e., at the school-by-year-by-subject level. This very granular level of fixed effects can be thought of as capturing teacher-level inputs because, in most schools and grades, a single teacher would teach a particular subject. The estimates that include additional subject-FX do not differ from the estimates that rely on school-level FX. This may indicate that individual teachers are not significantly affecting the home broadband effect.

³⁰ Note that we also explore potential non-linearity in the effect of available broadband speed. We split the sample into quartiles based on the distance to the LE station on the slow side of each boundary. Unfortunately, other observable characteristics also differ across these quartiles. As a consequence, we could not generate credible estimates that identify non-linearity net of heterogeneity using this approach.

Table 4

Fuzzy SRD estimates: Impact of exchange proximity on student performance by school technology.

	Quantile in school distance to LE				Early adopter	Late adopter
	(1)	(2)	(3)	(4)	(5)	(6)
Proximity to LE (negative distance)	0.00203** (0.00102)	−0.00009 (0.00098)	0.00077 (0.00105)	0.00065 (0.00081)	0.00152** (0.00073)	0.00101 (0.00062)
Distance LE to school (mts)	1–879	879–1354	1354–1912	1912–4716		
Months since upgrade					> 60	0–56
Observations	45,312	44,602	45,162	44,921	58,168	92,474

Notes: The table shows the fuzzy SRD nonparametric estimates for average test scores on the KS3 using the specification from column 4 of Table 3. Columns 1 to 4 split the sample into quartiles based on the school distance to the local exchange (LE). Columns 5 and 6 split the sample depending on the LE ADSL upgrade date for the school. Standard errors clustered at the segment-by-year level are reported in parentheses. ** $p < 0.05$.

Table 5

Impact of exchange proximity on student performance by student characteristics.

	Student Gender		KS1 Test Score		Ethnicity		FSM Eligibility	
	Boys	Girls	High	Low	White	Nonwhite	Yes	No
	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Proximity to LE (negative distance)	0.00098 (0.00067)	0.00125* (0.00068)	0.00156** (0.00068)	0.00096 (0.00066)	0.00114** (0.00051)	0.00211 (0.00133)	0.00073 (0.00149)	0.00132*** (0.00050)
Observations	91,106	91,856	89,229	93,763	144,194	38,443	27,496	155,252

Notes: The table shows the fuzzy SRD nonparametric estimates for average test scores on the KS3 using the specification from column 4 of Table 3 for different subgroups of students. FSM stands for Free School Meal. Each coefficient comes from a separate regression, where the running variable is the residential proximity to the LE station boundary segment. The window size for the residential distance to the LE station boundary segment is ± 300 meters, and only boundaries where the average exchange proximity jumps by at least 300 meters are included. Standard errors clustered at the segment-by-year level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

maths and science. This is because, unlike in the KS3 exam, students can choose additional exams, such as in “further maths”. We used the resulting grades to generate an overall percentile grade for each subject, following Lavy et al. (2012). Another reason that the resulting test scores are less precise compared to the finely-measured age-14 KS3 scores is that the individual exams are graded on a coarse letter-scale.

Table A.10 shows the resulting fuzzy-RDD estimates. We still find statistically significant effects on average scores, comparable in size to the baseline KS3 effects, but less precisely estimated. The subject-specific results are in line with those of KS3. These results show that the effect of broadband speed persists beyond KS3, and does not fade out two years later.

5. Broadband internet speed and student test scores

Up until this point, we have focused on using the fuzzy SRD to obtain a causal estimate of the effect of distance from the LE station on test scores, that is, β from Eq. (1). This corresponds to the estimation of LATE. However, we are ultimately interested in the relation between broadband speed and test scores. Unfortunately, postcode-level speed data for our study period does not exist. We use out-of-sample-period data to reverse engineer the distance-speed relation present during our study period.

To investigate this relation, we combine the linked postcode-to-exchange station telecom network data with data from the UK's telecommunication regulator (Ofcom) on the average internet speed experienced by households in 2012 and 2013. These data are explained in Section 2.2.3. The Ofcom dataset is based on the speeds of broadband connections operated by the main operators.³¹ To replicate average internet broadband speeds that are plausible for our period of analysis (2005–2010), we use only postcodes with average measure speeds

of up to 10 Mbit/s per second, thus excluding superfast broadband connections, which were not available.

The average internet speed in our resulting dataset is 5.5 Mbit/s and not far from UK consumers' actual average download speeds of 4.7 Mbit/s in 2008 (Ofcom, 2009b).³² The dataset provides us with the average speed (modem sync speeds) recorded in speed tests across individual households, which we can geolocate at the level of their full residential postcode. We use this information to link the speed microdata to the telecom network database discussed above to estimate the relationship between residential exchange proximity and available internet speed using the fuzzy SRD design.³³

Table 6 shows the fuzzy SRD estimates of the effect of exchange proximity on postcode average (download) speed. These estimates are based on an identical empirical strategy as the one used to derive the effects of distance on test scores, as discussed in Section 3.3. Various specifications that include different sets of technological controls result in very similar estimates based on the boundary jumps: for each additional 100 meters in distance to the connected exchange, speed decreases by 0.089 Mbit/s. This coefficient is statistically significant at the 1% level. This provides a causal estimate of the impact of proximity to the LE station on average speed, which we label β_{Speed} . Reassuringly,

³² Ofcom (2009b) reports that consumers living in urban areas received average download speeds of 4.3 Mbit/s in comparison with 3.3 Mbit/s among those living in rural areas. Average internet speed varies greatly from country to country. The highest averages can be observed in Asian countries (e.g., South Korea, Hong Kong, Singapore, and Japan) and Scandinavian countries (e.g., Norway, Sweden, Finland, and Denmark). See Inc. (2015).

³³ After 2008, the broadband internet infrastructure in the UK was updated to significantly less distance-sensitive technologies, such as coaxial cables or fibers. Due to this feature of our dataset, we regard our estimates of the effect of crossing the invisible catchment area boundary from the Slow to the Fast side on the available internet connection speed as a lower bound (for the test score estimation period). However, in 2012, 75% of broadband subscriptions were still for ADSL technologies, which is sensitive to spatial decay.

³¹ This dataset includes over 13 million connections in 2012 and over 19 million connections in 2013. Annexe 1 of the respective Ofcom infrastructure reports provides detailed descriptions (Ofcom, 2012; Ofcom, 2013).

Table 6
Fuzzy SRD estimates: Impact of exchange proximity on average broadband speed.

Dependent variable: Average internet speed (Mbit/s)			
	(1)	(2)	(3)
Proximity to LE (negative distance)	0.00080*** (0.00015)	0.00083*** (0.00016)	0.00089*** (0.00018)
Months since ADSL Upgrade			0.0002 (0.004)
Log Number of Premises			0.167* (0.085)
Log Area Square Meters			−0.190** (0.087)
Distance Bins	Yes	Yes	Yes
Area & Amenities Controls	No	Yes	Yes
Average internet speed (Mbit/s)	5.5	5.5	5.5
SD internet speed (Mbit/s)	2.7	2.7	2.7
Observations	20,274	20,274	20,274

Notes: The table shows the fuzzy SRD nonparametric estimates for the effect of exchange proximity on average internet speed (Mbit/s). The variable “Months since ADSL Upgrade” refers to the number of months since an ADSL upgrade occurred in the LE station catchment area. The variable “Log Number of Premises” refers to the logarithm of the number of premises in the LE station catchment area. The variable “Log Area Square Meters” refers to the logarithm of the square meters of the LE station catchment area. Each coefficient is from a separate regression, where the running variable is the residential proximity to the LE station boundary segment. The window size for the residential distance to the LE station boundary is ± 300 meters. Standard errors clustered at the segment-by-year level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

this speed-distance decay estimate is very similar to that obtained by Ahlfeldt et al. (2017) for ADSL technologies in 2009 and 2010.³⁴

By combining the estimate of the impact of proximity to the LE on test scores from Eq. (1), $\hat{\beta}$, with $\hat{\beta}_{Speed}$, we can compute a Wald estimate for the effect of available speed on test scores, ω :

$$\omega = \frac{\beta}{\beta_{Speed}} \quad (6)$$

Applying Eq. (6) to compute ω yields an effect of $\hat{\omega} = 1.37$. This implies that for each increase in broadband speed of one Mbit/s (corresponding to an increase of 20% with respect to the baseline average speed) test scores increase on average by 1.37 percentile points. This average effect is equivalent to approximately 5% of a standard deviation in the national test score distribution. The standard error of $\hat{\omega}$ is obtained using the Delta method and estimated with high precision (i.e., the p -value is < 0.01).³⁵ Even ω , which is the policy-relevant parameter, can still be interpreted as a reduced-form effect, as it reflects the effect of broadband availability – and not usage – on test scores.³⁶

³⁴ Ahlfeldt et al. (2017) use broadband speed tests carried out by individuals in 2009 and 2010 from a private company to estimate the speed-distance relationship for different internet technologies. The estimates of Ahlfeldt et al. (2017) imply that actual download speed decreases 0.085 Mbit/s for each additional 100 m further from the LE station for ADSL technologies (for distances ranging 1000–3000 meters from the LE). In addition, this estimate is not dissimilar to estimates provided by Ofcom using survey data in 2009 (Ofcom, 2009b).

³⁵ We use the test statistics:

$$\frac{\hat{\beta}}{\hat{\beta}_{Speed}} \sim N(0, 1) \quad (7)$$

Assuming that our estimates are independent, we can compute:

$$VAR\left(\frac{\hat{\beta}}{\hat{\beta}_{Speed}}\right) = \frac{1}{(\hat{\beta}_{Speed})^2} * (VAR(\hat{\beta}) + \left(\frac{\hat{\beta}}{\hat{\beta}_{Speed}}\right)^2 * VAR(\hat{\beta}_{Speed})) \quad (8)$$

³⁶ One of the channels through which we can surmise that broadband speed affects test scores is via broadband subscriptions take-up if it was unbalanced across boundary-segments. Using Ofcom postcode-level data for 2013, we test whether the effect of distance to LE on postcode take-up rates (the number of active broadband lines divided by the number of premises) changes across boundaries. We find precisely estimated zeros.

Notice we can also express the effect size in terms of standard deviations in broadband speed. The standard deviation of broadband speed is 2.7 Mbit/s. Our headline estimate implies that an increase in broadband speed of one standard deviation improves test scores by 13.5% of a standard deviation.

Finally, to assess the magnitude of the effect of broadband usage on test scores, we complement this Wald estimate with survey data evidence on broadband usage and information about the extent to which students used the internet for school work during our study period. Using data from Oxford Internet Surveys (OxIS) over several years, we compute broadband take-up for the group of interest in our sample years. We exploit this database because it allows us to be precise in the definition of the relevant population under study: English households with home broadband that had children aged between 14 and 17 years old.³⁷ We find that the weighted average of broadband usage among these families between 2005 and 2009 is 69.5%. These figures square with the finding reported by Livingstone and Bober (2005) that the vast majority of children used the internet at home, and most of them devoted their internet use to doing work for school or college. These authors' figures show that in the UK, 84% of 9- to 19-year-olds used the internet daily or weekly in 2005. Among those, 90% declared that they used the internet to do work for school or college.

Using these calculations, which refer to the exact age group that we study, scaling our Wald estimate by broadband take-up shows that for each increase in broadband speed of one Mbit/s, test scores increase on average by 1.97 percentile points. This increase is equivalent to approximately 7% of a standard deviation in the national test score distribution. Our headline estimate thus implies that a one-standard-deviation increase in broadband speed improves test scores by 19% of a standard deviation in test scores. This can be interpreted as the cumulative effect of having faster internet speed over six years on average. Assuming the effects are linear over time, an increase in broadband speed of one standard deviation is equivalent to an increase of 2.25% of a standard deviation in the national test score distribution (3.15% if we scale our estimate by broadband take-up) per year of exposure to faster internet speed.

³⁷ More aggregated datasets used in other studies, such as Eurostat or Ofcom data, produced similar numbers. The drawback of using these databases is that they, unlike OxIS, do not allow us to refine our calculations to the specific population of interest.

These effects are now directly comparable to the rest of the education literature and are not small in relative terms. For instance, teachers have been identified as one of the most important factors in test performance, with a one standard deviation increase in teaching quality improving test scores by 0.1–0.2 standard deviations (Chetty et al., 2014; Rivkin et al., 2005; Slater et al., 2012). Peer effects are smaller, at between 0.01 and 0.08 standard deviations (e.g., in the English context Gibbons & Telhaj, 2016; Lavy et al., 2012), and various dimensions of neighborhood quality have even smaller effects on test scores (e.g., Gibbons et al., 2013, 2017; Jacob, 2004; Sanbonmatsu et al., 2006). Therefore, in the context of the education literature, our estimates represent medium-sized effects.

6. Mechanisms - survey data evidence on internet use

The theoretical time taken to perform various online activities changes substantially with the internet connection speed (OfCom, 2010).³⁸ Faster internet speed increases the amount of information obtainable from the web or shared via the network per unit of time, affecting education performance in two ways. First, students might become more productive per unit of time invested in studying if online learning resources enhance learning. For this to be feasible, students need to have access to relevant study productivity-improving online tools, be aware of them, and use them appropriately. Between 2005 and 2008/10, multiple online resources offering educational support became available to teenagers. For instance, *YouTube* was launched in 2005, and soon after, educational institutions and individuals began uploading educational content with keen interest from the platform to help educators. Multiple popular education-oriented channels existed in this period (e.g., *Khan Academy*, *Smarter Every Day*, *TED Talks*).³⁹ On the other hand, with better connections, the possibility of engaging in social media and other distracting activities increases, potentially decreasing the amount of time spent doing schoolwork. Students could also make use of early social media tools (such as *Microsoft Messenger*) to share information and study together or visit sites such as *Sparknotes* to download essay-writing content. In fact, amongst the list of most viewed websites at the time were *Google*, *MSN Windows*, *Wikipedia* or *YouTube* (OfCom, 2007).

During our period of study, English children had access to and used the internet from a computer at home. Livingstone and Bober (2005) shows that in autumn 2004, 75% children aged 9–19 years old had access to the internet at home. OfCom (2006, 2008) provide similar figures for 2005 and 2007, with 68% and 77% of children aged 12–15 having access to the internet at home, respectively. According to OfCom (2008), the internet consumption for children aged 12–15 at home doubled between 2005 and 2007, from 7.1 to 13.8 average weekly hours. Hence, students were accessing to the internet at home, where they could use study-enhancing online tools. Parents perceive children's internet use with concern however, with the vast majority of

parents (75%) having rules in the household about access and internet content (Livingstone & Bober, 2005).

In this section we use additional data from the BHPS to explore the mechanisms driving our net results. To ensure that our results are not driven by changes in the extensive margin, e.g., households subscribing to faster broadband internet services more on the fast than the slow sides, we run additional analysis using the telecom data from Section 5 and BHPS data at the household level. These results are shown in Table A.13.

Columns (1) and (2) of Table A.13 report coefficients from the same specification of column (3) of Table 6, but with different outcomes. These are the log number of active broadband connections in a given postcode and the broadband penetration rate (number of active subscriptions over the number of residential addresses in the postcode, i.e., take-up rates). As explained in Section 5, this data comes from 2012 and 2013 postcode cross sections, but we trim the sample in order to make it representative of the connections and speeds available in 2007/2008. The first thing to notice is that the broadband penetration rate average in our dataset is 58%, which is extremely similar to 56.5%, the value of broadband penetration reported in Nardotto et al. (2015) using the actual number of connections in that year in 2007. This fact reassures us that our analysis is robust and representative for the period 2007/2008. The first two columns of the table show no statistical differences in the outcomes across fast-slow sides, and the coefficients are very close to zero.

To further confirm this result, as explained in Section 3.5, we use the location of the households at the LSOA level and adapt our empirical strategy to explore differences in ICT endowments across fast/slow borders. These coefficients are shown in columns (3)–(5) of Table A.13. For the year 2007 (wave Q), the survey provides information at the household level on computer ownership and internet/broadband subscriptions. We restrict our sample to households living within 1000 m of a LE catchment area boundary, and with jumps over 300 m, and estimate a reduced-form version of the main results empirical model. The regressions include household and individual level controls from BHPS as well as the same area and running variable controls as the main results estimations. Table A.11 show summary statistics for the key variables. ICT at home endowments are tested using information at the household level, as this information is collected with respect to the entire home. In the estimating sample, the rates of computer ownership are almost 80% on average, while around 72% of households and almost 65% have internet and broadband connections (a rate a bit higher than in the telecom data). The results in the table show no statistical differences in ICT endowments at home between the slow and fast sides of the boundaries. These results, together with the ones from the first two columns, are reassuring as they confirm that our paper estimates changes in the speed of the broadband connection (the intensive margin) and not simply subscriptions to high-speed internet (the extensive margin).

The BHPS survey also provides, for the year 2007, information on personal computer use for individuals in households for a set of different alternatives. Using the sample as for the household regressions, and further restricting to households where a personal computer is present, we explore the mechanisms driving our results by means of testing if our age group of interests, teenagers, uses computers for education more than the control group (non-teenagers), and if there are differences between fast and slow sides. Table A.12 shows summary statistics for the key variables for the individual sample.

In Table 7 we dig into the mechanisms driving our results, and estimate the specifications (5) described in Section 3.5. The table displays the effects on different margins of computer use at home at the individual level for respondents of different ages. The overall data comprises individuals from age 15 to 65, and we generate indicator variables for the youngest age group 15–17 and for young adults aged

³⁸ For instance, downloading a 250 Kbit/s web page takes 1 and 0.3 s with a provider of 2 Mbit/s and 8 Mbit/s, respectively. Besides, this difference grows exponentially with the online activity provided. Downloading a DVD-quality film (4 GB) takes 4 h and 48 min and 1 h and 11 min with a provider of 2 Mbit/s and 8 Mbit/s, respectively. See Figure 4.1 of OfCom (2010).

³⁹ The founder of the *Khan Academy* started posting videos in 2006 and created the academy channel in 2008. His videos became very popular, attracting tens of thousands of viewers every month. *Smarter Every Day* started its educational videos in 2007, and *TED Talks* began sharing talks in 2006. The demand for educational videos led to the creation of *YouTube EDU* in 2009 as a repository for its educational content. The number of articles in encyclopedia-type sites, such as *Wikipedia*, also increased exponentially since 2004. This free online encyclopedia has been found to be a relevant tool to improve learning (Derksen et al., 2022).

Table 7

Mechanisms: Individual computer uses for teenagers and young adults.

	Age Group 15–17				Age Group 18–24			
	(1) <i>Work</i>	(2) <i>Educ</i>	(3) <i>Games</i>	(4) <i>Hobbies</i>	(5) <i>Work</i>	(6) <i>Educ</i>	(7) <i>Games</i>	(8) <i>Hobbies</i>
Fast	0.016 (0.499)	−0.473 (0.496)	−0.875 (0.628)	−0.876 (0.695)	0.081 (0.515)	−0.691 (0.464)	−0.756 (0.606)	−0.619 (0.681)
AgeGroup	−0.042 (0.059)	0.318*** (0.119)	0.046 (0.095)	0.067 (0.125)	−0.043 (0.063)	0.251*** (0.064)	0.157* (0.093)	0.101 (0.090)
AgeGroup * Fast	−0.092 (0.083)	0.289** (0.120)	−0.034 (0.155)	0.015 (0.169)	0.001 (0.069)	0.074 (0.075)	−0.027 (0.091)	−0.134 (0.091)
Observations	1,102	1,102	1,102	1,102	1,051	1,051	1,051	1,051

Notes: The table shows the SRD nonparametric estimates for the individual computer usages for the specification (5) using wave Q (2007) data from the BHPS. Columns report different specifications. Each column estimates a separate regression, where the running variable is the residential proximity to the LE station boundary. The sample is restricted to households with home computers, and conditions on having broadband connections at home and individuals ever using the computer to access the internet from home. The window size for the residential distance to the LE station boundary is ± 1000 meters. We only include boundaries where the distance jumps by at least 300 meters. Standard errors clustered at the boundary level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$.

18–24.⁴⁰ We explore the use of personal computers (PC) for four different uses: paid work, educational work, playing games and hobbies. We restrict the sample to individuals in households which own personal computers, and controls for having a broadband connection at home and using the PC to access the internet. Cols (1) to (4) display the results for the teenage group (15–17), where the control group is everyone aged 18 and older. Cols (5) to (8) explore the effects for young adults – many of them in tertiary education – and we exclude minors here, so the control group is aged 25 and older (hence the smaller number of observations).

The first row shows the baseline (*Fast*), with no significant effects for being on the fast side for that particular use. This is the case for both teenagers and young adults. Again, these results are reassuring as they show balanced use for both age groups in all the PC usages, between the fast and slow sides. The second coefficient (*AgeGroup*) shows the differential use of a particular age group: the table shows that both teenagers and young adults use the PC more for educational work than their respective comparison groups, and the effect is larger for teenagers. For young adults, we also see an effect on the use of PC for games, although less significant. Finally, the interaction term (*AgeGroup * Fast*) shows the additional effect of that usage in the fast side with respect to the slow side, for that age group. The key results are that for teenagers, this interaction is significant and quite large for educational work, but not for distracting activities (games or hobbies) or for paid work. For young adults, although we did see they use the PC for education work more than 25–65 individuals, there are no differences between the slow and fast sides. These results support the hypothesis that the mechanisms behind the positive effects on student test scores seem to be a more education-oriented use of the internet combined with limited use of distraction activities.

7. Conclusion

This paper investigates the impact of ICT quality at home on educational outcomes, related to human capital accumulation relevant to labor market performance. We use a fuzzy SRD approach to present estimates of the causal net effect of available broadband speed on the test scores of 14-year-old pupils in England. Our headline estimate implies that a one-standard-deviation increase in broadband speed improves test scores by 19%

Our strategy further allows us to estimate the causal effects of home broadband availability interacting with school technology. Here, we find that the positive effects of home broadband on human capital only emerge if the school also has access to faster broadband. Our results

highlight how school and home environments interact to generate the positive human capital effects of broadband use.

Although most of the programs focused on providing broadband for educational purposes have been targeted at schools⁴¹, the first set of policies that our results inform relates to improving investments in faster broadband network infrastructure for households. At the household level, reducing spatial differences in the availability of super-fast broadband infrastructure is frequently discussed as a desirable outcome of policies implemented across the world.⁴² Governments have used the argument that investing public funds into infrastructure can boost firm performance and employment, but such arguments have rarely been applied to education-related policies (SQW, 2014). This is particularly salient in the UK with its recent ultrafast fiber broadband rollout, where major political concerns exist about rural areas being left behind and missing growth opportunities due to a lack of private ICT investment incentives (see, for example, DCMS, 2011; DDCMS, 2018).

In terms of the students benefiting, we find larger responses from students of more advantaged socio-economic backgrounds. Results do not fade out two years later, as we show significant effects on age-16 test scores for the same cohort of students. An examination of the mechanisms behind our findings shows that the beneficial effects of faster internet speed are driven by higher education-oriented use of the internet combined with no increase in the use of distraction activities. A direct implication from this heterogeneity in the effects is that better broadband connections at home contribute to educational inequalities. These results imply that home digital inequalities may widen the SES educational performance gap, which translates into persistent earning inequalities.

A second set of policies that our results speak to relate to improving the internet learning environment at home, particularly for low-SES households. Greater efforts are needed to enable students of lower ability or socio-economic background to also use broadband internet at home productively. Programs that focus on improving both access to fast connections and digital literacy with a special focus on low-income families, such as *BroadbandUSA*, *Internet Essentials* or the *ConnectHome* program, will likely be the most beneficial to improve student learning and reduce education inequalities.⁴³ Providing low-SES parents' with

⁴¹ For example, the British Department for Education committed in 2019 to provide gigabit (100 Mbit/s) capable speed broadband connections to over 100 rural primary schools.

⁴² For example, the EU's Gigabit Society, American NCTA 10G Gigabit Internet, and UK Gigabit-broadband programs.

⁴³ In addition, these types of interventions can have positive direct effects on improving low-income families' labor market prospects. For instance, Zuo (2021) shows that the *Internet Essentials* program increased employment rates and earnings for qualifying low-income families. Gallego et al. (2020) shows that children whose parents receive information about their internet use spend less time online.

⁴⁰ Unfortunately, data for younger children is not provided, but the youngest group at least partly overlaps with the age-16 KS4 sample, making this informative for the mechanisms at play.

more appropriate tools to understand the risks and benefits of access to the internet seems key to improving their ability to help their children take full advantage of the EduTech potential.

Our results speak to the recent evidence on the negative impacts of certain forms of technology use in the classroom, particularly mobile devices such as smartphones and tablets. In 2023, UNESCO (Antoninis et al., 2023) has advocated for banning smartphones in schools to reduce classroom disruptions and protect students from online harm. Banning smartphones has been found to significantly improve test scores (Beland & Murphy, 2016), as well as enhance students' psychological well-being and increase enrollment in academic high school tracks (Abrahamsson, 2024). Furthermore, Abrahamsson (2024) finds that these benefits are especially pronounced among girls from low-SES households. Similarly, studies show that tablet use in class can lead to a general decline in academic achievement (Dauphin, 2019). This growing body of evidence suggests that the use of technology at both school and home should be carefully evaluated to ensure that potential distractions do not undermine its capacity to enhance educational outcomes, particularly for students from disadvantaged backgrounds.

Our findings are even more pertinent in the context of the post COVID-19 pandemic, in which many countries closed schools for several months and home online learning became a major substitute for in-classroom teaching. The pandemic brought the consequences of unequal access to technology to the spotlight, as families with

fewer resources could not provide their children with the appropriate infrastructure to engage in digital school work (Andrew et al., 2020; Bacher-Hicks et al., 2021; Stantcheva, 2022). The results highlight the value added to broadband investments and the importance of ensuring universal access and digital skills to mitigate increases in inequalities in educational opportunity in times of regular schooling.

Last but not least, the heterogeneity in the effects of home broadband by school technology sends an important message to educational policy: to reap its full effectiveness, technology used at the school level ought to be used at home, as well. This finding of complementarity between school and home technology likely extends into future technologies and changes to the process of human capital accumulation, beyond the effects of broadband.

Declaration of competing interest

No financial or other interests to declare.

Appendix. Appendix figures and tables

See Figs. A.1–A.5 and Tables A.1–A.13.

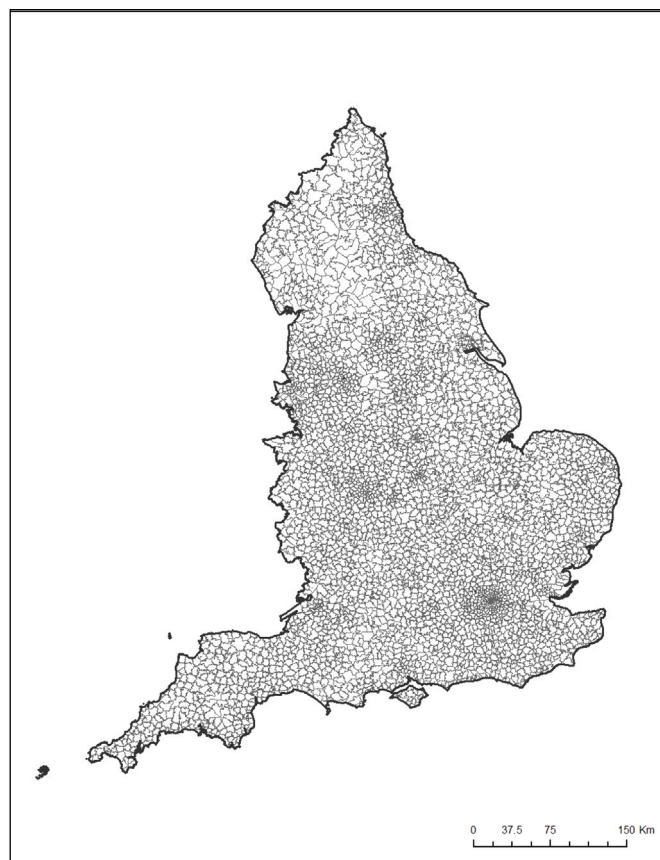


Fig. A.1. Telephone local exchange areas in England

Notes: There are 3,978 local exchange areas in England (of which 3,925 are completely contained in England). The average area is of 34 square kilometres, and serves an average of 5,830 premises (with 93% of them residential).

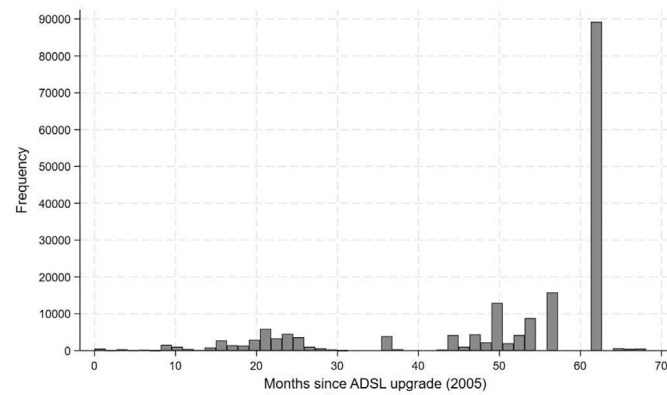


Fig. A.2. Distribution of months since DSL upgrade (in 2005).

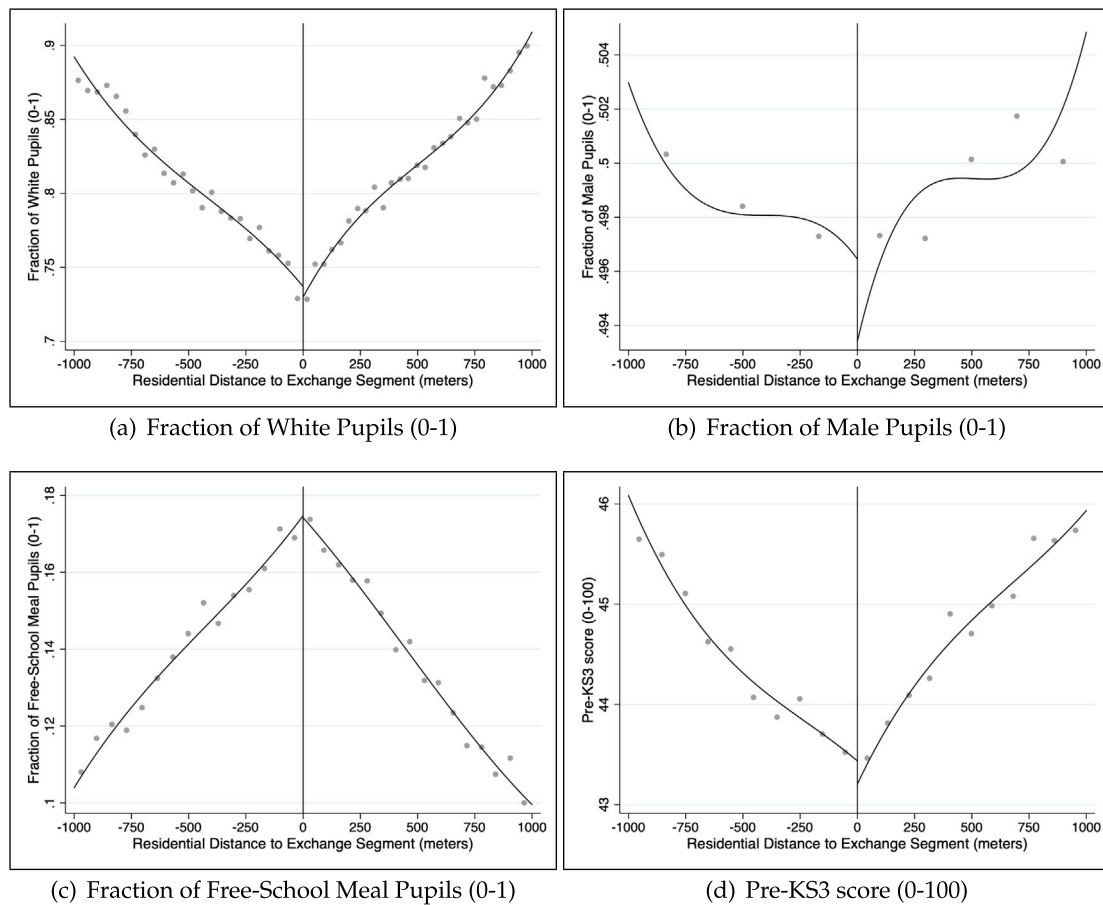


Fig. A.3. Students characteristics.

Notes: The dots represent the fraction of white pupils, the fraction of male pupils, the fraction of free-school meal pupils and the pre-internet score per interval of residential distance to the exchange segment boundaries. The solid lines are fitted values from a third order polynomial approximation, which is estimated separately on both sides of the cutoffs. “Residential Proximity to Exchange Segment (meters)” refers to the residential proximity to the exchange segment boundaries. Black vertical lines identify the LE station boundaries.

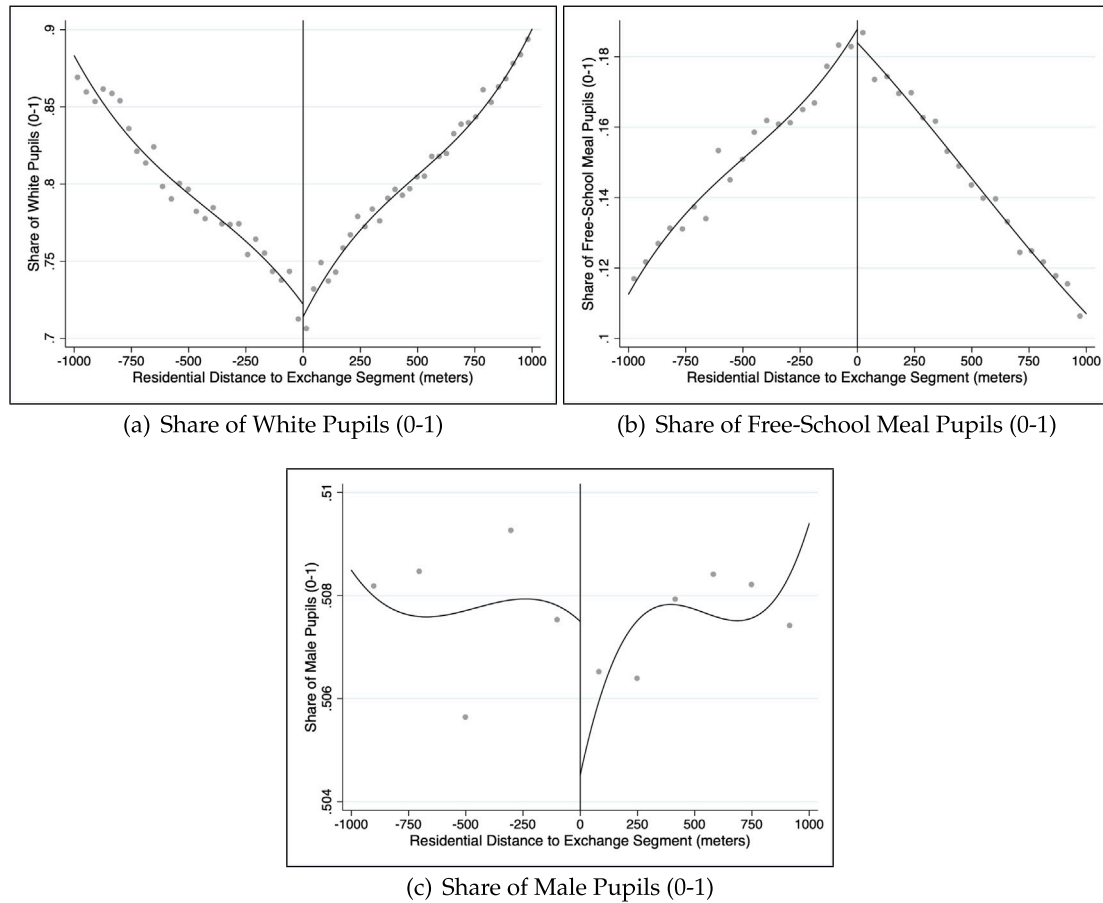


Fig. A.4. Density & area socio-economics.

Notes: The dots represent the share of white pupils, share of free-school meal and share of male pupils per interval of residential proximity to the exchange segment boundaries.. The solid lines are fitted values from a third order polynomial approximation, which is estimated separately on both sides of the cutoffs. “Residential Proximity to Exchange Segment (meters)” refers to the residential proximity to the exchange segment boundaries. Black vertical lines identify the LE station boundaries.

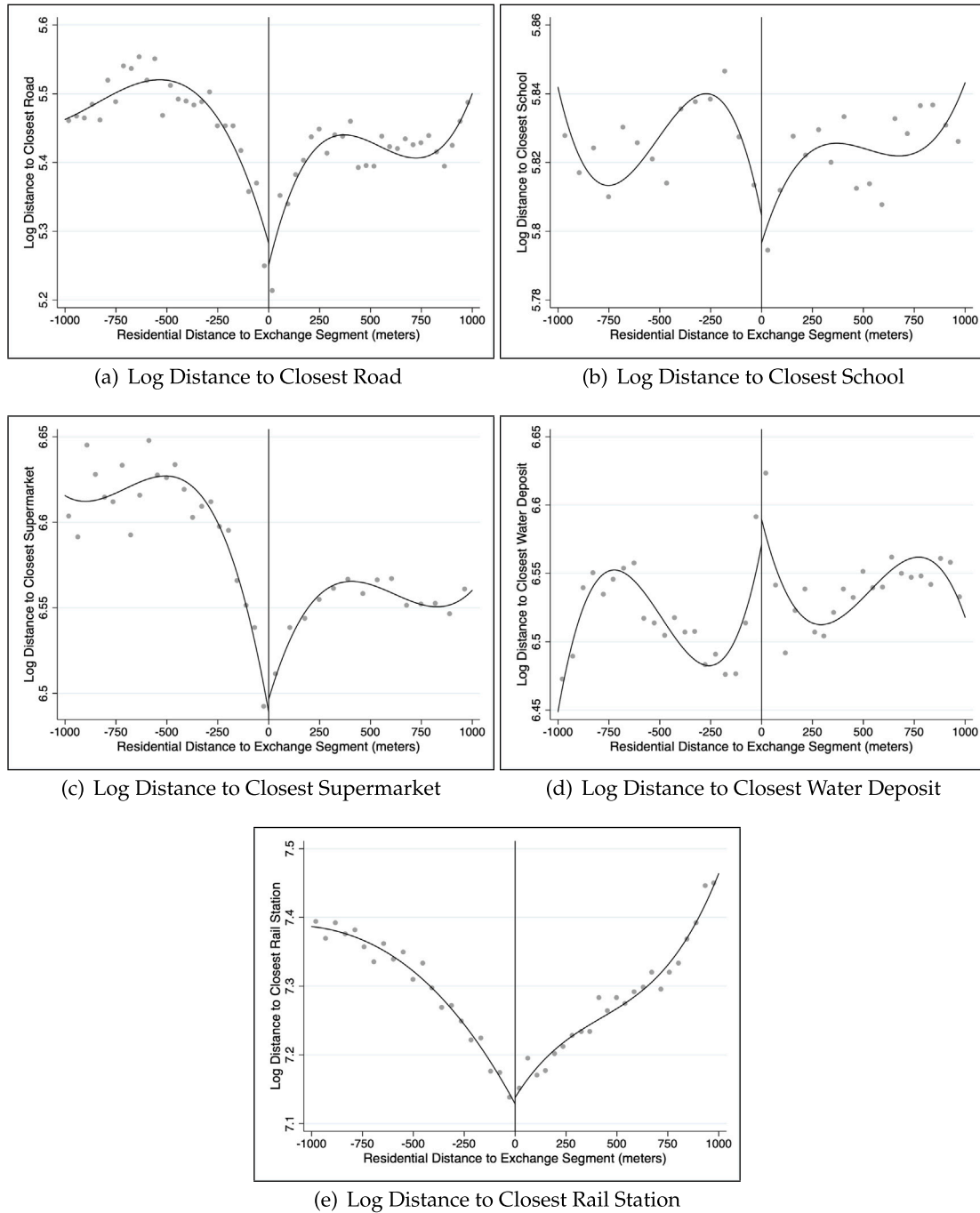


Fig. A.5. Amenities.

Notes: The dots represent the logarithm distance to closest road, school, supermarket, water deposit and rail station per interval of residential distance to the exchange segment boundaries. The solid lines are fitted values from a third order polynomial approximation, which is estimated separately on both sides of the cutoffs. "Residential Distance to Exchange Segment (meters)" refers to the residential distance to the exchange segment boundaries. Black vertical lines identify the LE station boundaries.

Table A.1
Summary of studies analyzing the effect of internet access on student achievement.

Study	At-home vs. at-school	Ext. vs. Int. margin	Positive vs. Negative eff.	Student age	Developed vs. Developing country	Years of data	Outcome variable
Panel A: Studies Analyzing the Effects of Internet Access on Student Performance							
Dettling et al. (2018)	H	Ext.	Positive	High School graduation	Developed (US)	1999–2008	College application, SAT
Malamud et al. (2019)	H	Ext.	Null results	Age 9–11 (Grades 3 to 5)	Developing (PE)	2011–2013	Math and reading achievement
Vigdor et al. (2014)	H	Ext.	Negative	Age 11–14 (Grades 5–8)	Developed: N.C.(US)	2000–2005	Math and reading achievement
Lakdawala et al. (2023)	S	Ext.	Positive	Age 7 (Grade 2)	Developing (PE)	2007–2016	Math and reading achievement
Belo et al. (2014)	S	Ext. & Int.	Negative	Age 12 (Grade 9)	Developed (PT)	2005–2009	Achievement
Belo et al. (2016)	S	Ext.	Positive	Age 6–12	Developed (PT)	2006–2009	Internet penetration
Hopkins B (2014)	S	Ext.	Positive/null (mix)	Grade 2 of primary school	Developing (PE)	2007–2011	Math and reading achievement
Panel B: Studies Analyzing the Effects of other Type of ICT on Student Performance							
Goolsbee and Guryan (2006)	S	Ext.	Null results	Prim. & Second. Educ.	Developed (CA, US)	1996–2000	Math and reading achievement
Machin et al. (2007)	S	Ext.	Positive	Age 11 (Grade 8)	Developed (GB)	1999–2003	Math and reading achievement
Angrist and Lavy (2002)	S	Ext.	Null	Elementary and Middle	Developed (IL)	1996	Test scores
Banerjee et al. (2007)	S	Ext.	Positive	Primary school (grade 4)	Developing (IN)	2002–2004	Math achievement
Barrera-Osorio and Linden (2009)	S	Ext.	Null (or little)	Grades 1–11	Developing (CO)	2006, 2008	Math and Spanish test scores
Barrow et al. (2009)	S	Ext.	Positive	High and middle schools	Developed (US)	2003–2005	Math achievement
Beuermann et al. (2015)	H	Ext.	Null	Grade 2	Developing (PE)	2011	Math achievement
Cristia et al. (2017)	H & S	Ext.	Null	Grades 2–6	Developing (PE)	2010	Math achievement
Derksen et al. (2022)	S	Ext.	Positive	Secondary school	Developing (MW)	2018	English & Biology achievement
Fairlie and Robinson (2013)	H	Ext.	Null	Grades 6–10	Developed (US)	2008–2010	Grades & test scores
Falck et al. (2018)	S	Ext.	Null (mix)	Grades 4 and 8	50+ countries	2011	Math and Science achievements
Malamud and Pop-Eleches (2011)	H	Ext.	Mix	Under 26	Developing (RO)	2009	Cognitive & school grades
Muralidharan et al. (2019)	S	Ext.	Positive	Middle school	Developing (IN)	2015	Math and Hindi achievement
Leuven et al. (2007)	S	Ext.	Negative	Grade 8	Developed (NL)	1998–2003	Test scores
Mo et al. (2014)	S	Ext.	Positive	Grades 3 and 5	Developing (CN)	2011–2012	Math scores
Rouse and Krueger (2004)	S	Ext.	Positive/Null	Grade 6	Developed (US)	2001–2002	Language and reading skills
Fuchs and Wossmann (2004)	H & S	Ext.	Negative & Null	15 years old	32 countries	2000	PISA scores
Cristia et al. (2012)	H & S	Ext.	Null	Primary school	Developing (PE)	2010–2012	Math and reading achievement
Suhr et al. (2010)	H & S	Ext.	Positive	Grades 4 and 5	Developed (US)	2004–2005	English achievement
Panel C: Studies Analyzing the Effects of Internet Access on Non-Education Outcomes							
Akerman et al. (2015)	F	Ext.	Mix		Developed (NO)	2000–2008	Wages
Czernich et al. (2011)		Ext.	Positive		OECD countries	1996–2007	Economic growth
Ahlfeldt et al. (2017)	H	Int.	Positive		Developed (GB)	1995–2010	Property value
Amaral-Garcia et al. (2019)	H	Ext.	Positive		Developed (GB)	2000–2011	Cesarean Sections
Bellou (2015)	H	Ext.	Negative		Developed (US)	1990–2006	Marriage rates
Falck et al. (2014)	H	Ext.	Negative		Developed (DE)	2005–2008	Voter turnout
Campante et al. (2018)	H	Ext.	Mix		Developed (IT)	1996–2013	Voter turnout, and other
Gavazza et al. (2019)	H	Ext.	Negative		Developed (GB)	2006–2010	Voter turnout, and other
Bhuller et al. (2013)	H	Ext.	Positive		Developed (NO)	2000–2008	Sex crimes
Geraci et al. (2022)	H	Ext.	Negative and Null		Developed (GB)	1997–2017	Civic and pol. participation
Johnson and Persico (2024)	H	Ext.	Positive		Developed (US)	2000–2008	Economic and mental health
Zuo (2021)	H	Ext.	Positive		Developed (US)	2012	Employment and earnings

Notes This table lists representative papers that investigate the relationship between internet access and student performance. The columns include: (a) at-home vs. at-school connectivity, (b) extensive vs. intensive margin, (c) positive vs. negative effects, (d) student age, (e) developed vs. developing countries, (f) years of data, and (g) outcome variable.

Table A.2

McCrary (2008) and Cattaneo et al. (2018) test for manipulation of the forcing variable for the different treatment subamples.

A. McCrary Test

Distance (m)	Log difference in frequency bins	Z-stat	Bandwidth	Bin size
250 m	.267 (0.206)	1.29	40.14	.556
300 m	-.024 (0.068)	.347	44.92	.604
350 m	-.043 (0.048)	.88	50.22	.65
400 m	-.038 (0.04)	.936	54.5	.69
450 m	-.038 (0.035)	1.07	59.69	.73
500 m	-.041 (0.031)	1.33	66.7	.77

B. Cattaneo et al. (2018): RD Manipulation Test using local polynomial density estimation:

Bias-corrected Density Estimate to the left of the cutoff	.00035 (0.000017)
Bias-corrected Density Estimate to the right of the cutoff	.00033 (0.000018)
T-test for bias-corrected density test	-0.7886
P-value for bias-corrected density test	0.4303

Notes: This table show the McCrary (2008) (Panel A) and Cattaneo et al. (2018) test for manipulation of the forcing variable. The McCrary (2008) test is performed separately for each treatment sample. The table columns of Panel A show the estimated discontinuity in the density function of the assignment variable at the threshold, its standard error (in parentheses), the associated z-statistic, the estimated optimal bandwidth, bin size and the number of observations. The optimal bandwidth and bin size are obtained using the selection procedure proposed by McCrary (2008). The table rows of Panel B present the bias corrected density estimate to the left and right of the LE boundaries, the t-test for the bias-corrected density test and the p-value of the test.

Table A.3

Sharp-SRD estimates: The impact exchange proximity on student performance.

	(1)	(2)	(3)	(4)	(5)
Effect of exchange proximity on national age-14 KS3 exam scores (in m)					
Proximity to LE	0.00043***	0.00034**	0.00027**	0.00025*	0.00024*
(negative distance)	(0.00014)	(0.00014)	(0.00013)	(0.00013)	(0.00013)
Student Pre-Internet Score KS1	Yes	Yes	Yes	Yes	Yes
Student Controls	No	Yes	Yes	Yes	Yes
Time-Variant Area & Amenities	No	No	Yes	Yes	Yes
Time-Invariant Area & Amenities	No	No	No	Yes	Yes
School Controls	No	No	No	No	Yes
Observations	183,892	183,892	183,892	183,892	183,892

Notes: The table shows the sharp SRD non-parametric estimates for test scores in KS3. Columns report different specifications. Each coefficient comes from a separate regression, where the running variable is the residential proximity to the LE station boundary. The window size for the residential proximity to the LE station boundary is ± 300 meters. We only include boundaries where the proximity jumps by at least 300 meters. Standard errors clustered at the segment-by-year level are reported in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.4

Sample choice based on distance to segment-boundary.

		(1)	(2)	(3)	(4)	(5)	(6)	(7)
Effect of exchange proximity on national age-14 KS3 exam scores								
		Distance (m)						
		100	150	200	300	500	750	1000
Proximity to LE	0.00157**	0.00154**	0.00134***	0.00122**	0.00092**	0.00101**	0.00100**	
(negative distance)		(0.00066)	(0.00060)	(0.00051)	(0.00047)	(0.00045)	(0.00043)	(0.00043)
Observations		49,303	85,182	119,55	183,892	305,404	431,709	530,773

Notes: The table shows the fuzzy SRD non-parametric estimates for the average proximity to LE station (Panel A) and test scores in KS3 (Panel B) with different samples based on the distance to segment-boundary. Columns report different specifications. Each coefficient comes from a separate regression, where the running variable is the residential distance to the LE station boundary. We only include boundaries where the distance jumps by at least 300 meters. Standard errors clustered at the segment-by-year level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.5

Sample choice based on magnitude of jump across segment-boundary.

	(1)	(2)	(3)	(4)	(5)	(6)
Avg. jump in proximity to connected exchange when crossing boundary to the Fast side (in m)						
Fast side	653.788***	731.578***	775.172***	816.649***	915.704***	999.089***
Estimates	(13.477)	(14.451)	(15.254)	(15.927)	(17.194)	(18.996)

Effect of exchange proximity on national age-14 KS3 exam scores (in percentile ranks)

	Minimum Boundary Jump (in m)					
	100	200	250	300	400	500
Proximity to LE	0.00064	0.00080*	0.00116**	0.00122**	0.00131***	0.00135***
(negative distance)	(0.00051)	(0.00049)	(0.00048)	(0.00047)	(0.00045)	(0.00046)
Observations	240,346	211,9396	196,315	183,892	158,022	134,078

Notes: The table shows the fuzzy SRD non-parametric estimates for the average proximity to LE station (Panel A) and test scores in KS3 (Panel B) with different samples based on the minimum boundary jump across the segment-boundary. Columns report different specifications. Each coefficient comes from a separate regression, where the running variable is the residential distance to the LE station boundary. The window size for the residential distance to the LE station boundary is ± 300 meters. Standard errors clustered at the segment-by-year level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.6

Fuzzy SRD estimates: Impact of exchange proximity on student performance with school ICT controls.

	(1)	(2)	(3)	(4)	(5)
Proximity to LE	0.00122**	0.00123***	0.00103***	0.00092**	0.00089*
(negative distance)	(0.00047)	(0.00047)	(0.00047)	(0.00046)	(0.00047)
School Proximity to LE Station		0.00050***	0.00058***		
		(0.00010)	(0.00013)		
School is early ADSL Adopter		0.75324***			
		(0.25776)			
School LE Station Fixed Effect	No	No	Yes	Yes	Yes
School Fixed Effect	No	No	No	Yes	Yes
School-by-year Fixed Effect	No	No	No	No	Yes
Observations	183,892	183,712	183,712	183,638	183,598

Notes: The table shows the fuzzy SRD nonparametric estimates for average test scores on the KS3. Columns report different specifications. Each coefficient comes from a separate regression, where the running variable is the residential proximity to the LE station boundary. The window size for the residential proximity to the LE station boundary is ± 300 meters. We only include boundaries where the proximity jumps by at least 300 meters. Standard errors clustered at the segment-by-year level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.7

Fuzzy-SRD estimates: The impact of crossing the broadband boundary on student performance by clustering choices for standard errors.

	(1)	(2)	(3)	(4)	(5)
Proximity to LE (negative distance)	0.00122** (0.00047)	0.00122** (0.00032)	0.00122** (0.00052)	0.00122*** (0.00047)	0.00122** (0.00051)
Segment-Year Level	Yes	No	No	No	No
Segment and Year Level	No	Yes	No	No	No
Segment Level	No	No	Yes	No	No
Boundary-Year Level	No	No	No	Yes	No
Boundary Level	No	No	No	No	Yes
Observations	183,892	183,892	183,892	183,892	183,892

Notes: The table shows the fuzzy SRD non-parametric estimates for test scores in KS3 by different clustering choice for standard errors. Columns report different specifications. Each coefficient comes from a separate regression, where the running variable is the residential proximity to the LE station boundary. The window size for the residential proximity to the LE station boundary is ± 300 meters. We only include boundaries where the proximity jumps by at least 300 meters. Standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.8

Fuzzy SRD estimates: Impact of exchange proximity on student performance by subject.

Dependent variable: National exam scores at age 14 (KS3)

		Average (1)	English (2)	Maths (3)	Science (4)
Proximity to LE (negative distance)	0.00122**	0.00141*** (0.00047)	0.00075 (0.00051)	0.00117** (0.00048)	(0.00050)
Observations		183,892	183,892	183,892	183,892

Notes: The table shows the fuzzy SRD nonparametric estimates for average test scores on the KS3 by subject using the specification from column 4 of Table 3. Each coefficient comes from a separate regression, where the running variable is the residential distance to the LE station boundary segment. The window size for the residential distance to the LE station boundary segment is ± 300 meters, and only boundaries where the exchange distance jumps by at least 300 meters are included. Standard errors clustered at the segment-by-year level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.9

Impact of exchange proximity on student performance by student characteristics.

Dependent variable: National exam scores at age 14 (KS3)

	KS1 High Non-FSME (1)	KS1 Low Non-FSME (2)	KS1 High FSME (3)	KS1 Low FSME (4)
Proximity to LE (negative distance)	0.00148** (0.00069)	0.00123* (0.00074)	0.00395 (0.00366)	-0.00080 (0.00169)
Observations	80,377	73,855	7,374	18,659

Notes: The table shows the fuzzy SRD nonparametric estimates for average test scores on the KS3 using the specification from column 4 of Table 3 for different subgroups of students. FSME stands for Free School Meal Eligible. Each coefficient comes from a separate regression, where the running variable is the residential proximity to the LE station boundary segment. The window size for the residential distance to the LE station boundary segment is ± 300 meters, and only boundaries where the average exchange proximity jumps by at least 300 meters are included. Standard errors clustered at the segment-by-year level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.10

Impact of exchange proximity on age-16 KS4 student performance.

Dependent variable: National KS4 exam scores at age 16

		Average (1)	English (2)	Maths (3)	Science (4)
Proximity to LE (negative distance)	0.00111*	0.00106* (0.00059)	0.00070 (0.00059)	0.00108* (0.00060)	(0.00061)
Observations		138,680	138,680	138,680	138,680

Notes: The table shows the fuzzy SRD nonparametric estimates for average test scores on the KS4 overall, and by subject. Each coefficient comes from a separate regression, where the running variable is the residential distance to the LE station boundary segment. The window size for the residential distance to the LE station boundary segment is ± 300 meters, and only boundaries where the exchange distance jumps by at least 300 meters are included. Standard errors clustered at the segment-by-year level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.11

Summary statistics BHPS households sample - Wave Q 2007.

Variable	Observations	Mean	Std. dev.	Min	Max
Home computer in accommodation	540	0.798	0.402	0	1
Have access to the internet from home	540	0.720	0.449	0	1
Broadband connection from home computer	540	0.646	0.479	0	1
Fast side	540	0.507	0.500	0	1
Proximity to LE	540	-1495.631	861.909	-6487.107	-73.502
Male (head-of-household)	540	0.644	0.479	0	1
Age (head-of-household)	540	44.719	12.453	18	65
Children in household	540	0.309	0.463	0	1
SES group 1	540	0.215	0.411	0	1
SES group 2	540	0.339	0.474	0	1
SES group 3	540	0.180	0.384	0	1
Single parent	540	0.057	0.233	0	1
Home owner	540	0.735	0.442	0	1

Notes: Sample is restricted to households where we could calculate the telecom variables reliably at the LSOA level, to those within 1000 m of a boundary and in boundaries with at least 300 m cable jump.

Table A.12

Summary statistics BHPS individuals sample - Wave Q 2007.

Variable	Observations	Mean	Std. dev.	Min	Max
Home computer in accommodation	1102	1.000	0.000	0	1
Have access to the internet from home	1102	0.918	0.274	0	1
Broadband connection from home computer	1102	0.852	0.355	0	1
Fast side	1102	0.487	0.500	0	1
Proximity to LE (negative distance)	1102	-1556.305	898.571	-6487.107	-55.248
Age 15–17	1102	0.044	0.206	0	1
Age 18–24	1102	0.145	0.352	0	1
Age 25–65	1102	0.810	0.392	0	1
Male	1102	0.478	0.500	0	1
Age	1102	39.683	13.829	15	65
Children in hhld	1102	0.324	0.468	0	1
SES group 1	1102	0.207	0.405	0	1
SES group 2	1102	0.345	0.476	0	1
SES group 3	1102	0.198	0.399	0	1
Single parent	1102	0.045	0.208	0	1
Home owner	1102	0.785	0.411	0	1

Ever use the computer to...

Connect to internet	1102	0.782	0.413	0	1
Educational work	1102	0.208	0.406	0	1
Paid work	1102	0.205	0.404	0	1
Playing games	1102	0.342	0.475	0	1
Hobbies	1102	0.352	0.478	0	1

Notes: Sample is restricted to individuals households where we could calculate the telecom variables reliably at the LSOA level, to those within 1000 m of a boundary and in boundaries with at least 300 m cable jump and in households where a computer was present.

Table A.13

Mechanisms: Broadband penetration and ICT endowments across boundaries.

	Telecom postcode data		BHPS Households 2007		
	(1) Log number connections	(2) Broadband penetration	(3) Computer at home	(4) Internet at home	(5) Broadband at home
Proximity to LE (negative distance)	−0.0046 (0.0190)	−0.0209 (0.0199)	0.098 (0.548)	0.318 (0.573)	−0.097 (0.653)
Months since ADSL Upgrade	−0.0012*** (0.0004)	−0.0010* [0.00050]			
Log Number of Premises	0.0117 (0.0082)	−0.0383*** (0.0097)			
Log Area Square Meters	−0.0082 (0.0082)	0.0164 (0.0175)			
Observations	15,319	20,274	540	540	540
Mean outcome	2.63	0.58	0.80	0.72	0.65

Notes: the table shows the SRD nonparametric estimates for different outcomes using telecom and BHPS data. Each column estimates a separate regression, where the running variable is the residential distance to the LE station's boundary. The first two columns test broadband penetration differences across boundaries using postcode-level telecom data. The last three columns test ICT at home variables using households observations. In columns 4 and 5 we condition on having a personal computer at home, on top of the rest of area, household and individual level controls. The window size for the residential distance to the LE station boundary is ± 1000 meters. We only include boundaries where the distance jumps by at least 300 meters. Standard errors clustered at the boundary level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$.

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